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MediSyn: A Synergistic multi-agent AI system for IPE in medical domain

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Abstract

The increasing complexity of modern healthcare demands collaborative, inter-professional approaches to ensure comprehensive patient care, yet current AI solutions based on general large language models (LLMs) often fall short in delivering nuanced, profession-specific advice. This project addresses the challenge of integrating Inter-Professional Education/Practice (IPE/IPP) into an AI-driven multi-agent system designed to generate treatment plans for real-life patient cases. We demonstrate that a baseline single LLM solution performs poorly when tasked with generating treatment plans. To overcome these limitations, we propose an innovative approach that simulates real-world healthcare team interactions by incorporating a multi-agent system that dynamically creates agents based on the professions represented in each case study. Rather than relying on a hardcoded set of agents, our system extracts team member roles from real-world inter-professional case studies and assigns each agent the domain-specific knowledge and responsibilities of its corresponding profession. This approach allows, for example, a nurse agent to provide patient care insights and an oncologist agent to deliver targeted cancer treatment advice, closely mirroring real-world inter-professional collaboration. Our experiments show that this specialized, collaborative framework significantly outperforms the general-purpose LLM, paving the way for improved, context-aware treatment planning in clinical settings.

Contents

1 Introduction

In modern healthcare, the complexity of patient needs requires collaborative approaches among diverse healthcare professionals. Interprofessional Education (IPE), as defined by the World Health Organization (WHO), occurs when “students from two or more professions learn about, from and with each other to enable effective collaboration and improve health outcomes” [1]. Building upon this educational foundation, Interprofessional Practice (IPP) involves “multiple health workers from different professional backgrounds working together with patients, families, carers, and communities to deliver the highest quality of care” [1]. These practices have been shown to enhance communication, reduce medical errors, and improve clinical effectiveness by aligning profession-specific expertise toward common goals.

With the emergence of LLMs in medical AI applications, there is growing interest in leveraging their capabilities to support or simulate such interprofessional reasoning. However, general-purpose LLMs often lack the role awareness and domain-specific context needed to replicate nuanced clinical decision-making, especially in multirole environments. These models typically produce generic treatment plans that may overlook profession-specific insights critical to comprehensive patient care.

In this work, we investigate how different LLM-based approaches perform in generating treatment plans for real-world IPE case studies. Specifically, we compare three paradigms: (1) a single general-purpose LLM, (2) a prompt-tuned LLM, and (3) MediSyn — a novel AI-driven multi-agent system that simulates an interprofessional care team by dynamically instantiating specialized agents for each role mentioned in a case. Rather than hardcoding professions, MediSyn parses the team members from the input and assigns each agent domain-specific responsibilities. For example, a nurse agent is fine-tuned to provide insights into patient care, while an oncologist agent delivers targeted cancer treatment recommendations.

Importantly, our goal is not to position MediSyn as a universally superior alternative, but rather to explore how such a system compares to more general-purpose LLM pipelines. We ask: Can simulating interprofessional dynamics through a role-specialized multi-agent framework provide advantages — or trade-offs — relative to current monolithic approaches?

Moreover, our extensive review of the current literature and AI applications in healthcare indicates that, although there are various multi-agent systems addressing tasks such as emergency response [2], interdisciplinary decision support [3], and cancer care information management [4], no existing system to our knowledge has been designed to tackle real IPE case studies using generative AI.

Through this work, we aim to provide empirical insights into the capabilities and limitations of these AI paradigms in replicating interprofessional treatment planning, and inform the design of future AI systems for interprofessional healthcare collaboration.

The remainder of this report is organized as follows: Section 2 reviews related work on AI-driven medical systems, multi-agent approaches, and LLMs in healthcare. Section 3 describes our methodology, including the dataset, preprocessing strategies, baseline generation pipelines, and the design of the MediSyn framework. Section 4 presents our experimental results comparing single LLM, prompt-tuned LLM, and multi-agent outputs across multiple evaluation metrics. Section 5 discusses key insights, limitations, and open challenges. Finally, Section 6 concludes the report and suggests directions for future work.

2 Related Work

The integration of large language models (LLMs) in healthcare has spurred growing interest in their use for tasks such as clinical decision support, treatment recommendation, and patient education [5–8]. While models like GPT-3 and GPT-4 have demonstrated impressive capabilities in understanding and generating medical language [9, 10], they often lack the role-specific contextual awareness necessary for profession-targeted treatment planning [11]. Their responses may be overly generic, missing the nuanced reasoning that emerges from collaborative healthcare practice involving diverse specialties.

To better address distributed reasoning, multi-agent systems (MAS) have been explored extensively in healthcare. MAS architectures simulate collaboration by assigning responsibilities to individual agents representing different medical roles [12–14]. Applications include chronic disease management, prehospital triage, cancer care coordination, and emergency response systems. These systems enable real-time communication, adaptive decision-making, and decentralized control—making them well-suited for modeling complex healthcare processes [13, 14].

The emergence of LLM-powered MAS frameworks bridges the gap between generative text reasoning and collaborative simulation. For instance, ClinicalAgent [15] demonstrates the use of GPT-4-based agents for simulating clinical trial coordination, where each agent is assigned a domain-specific function, such as patient recruitment or outcome analysis. Similarly, Han and Choi [2] developed an LLM-based multi-agent system for emergency triage using the Korean Triage and Acuity Scale (KTAS), dynamically orchestrating agents like triage nurse, physician, and coordinator for treatment planning. These systems highlight the growing feasibility of combining natural language understanding with agent-based role simulation in medicine.

Recent advancements have introduced sophisticated LLM-based multi-agent frameworks tailored for complex medical tasks. For instance, the KG4Diagnosis system integrates hierarchical agents with knowledge graphs to enhance diagnostic reasoning across various medical specialties [16]. Similarly, the MedAide framework employs specialized agents collaborating through retrieval-augmented generation to provide personalized healthcare services [17]. The MDAgents platform emphasizes adaptive collaboration among expert LLMs to tackle intricate medical decision-making processes [18]. Furthermore, the Multi-Agent Conversation (MAC) framework has demonstrated significant improvements in diagnostic accuracy by simulating collaborative discussions among agents [19]. These innovations underscore the potential of multi-agent LLM systems to replicate the collaborative dynamics of healthcare professionals, offering promising avenues for enhancing clinical decision support.

However, existing work has not yet addressed interprofessional education (IPE) or interprofessional practice (IPP) case studies using generative AI in a fully dynamic, role-adaptive way. Most prior systems rely on static agent configurations, do not reflect the diversity of real-world care teams, or are not designed for educational or collaborative training contexts [3, 11]. Our work builds upon these foundations by introducing MediSyn, an LLM-powered multi-agent system that dynamically extracts team roles from real IPE case studies and assigns them to profession-specialized agents. Rather than aiming to outperform prior methods, we explore how this novel architecture compares to single-LLM and prompt-tuned LLM baselines in generating treatment plans, particularly in scenarios requiring collaborative, role-sensitive reasoning.

3 Solution and Methodology

This section provides an overview of our methodology. Our approach is designed to evaluate the performance of a novel multi-agent AI system, MediSyn, for generating treatment plans in IPE healthcare scenarios. The methodology involves a structured pipeline, progressing from data extraction and preprocessing to various LLM-based generation approaches, culminating in the design and evaluation of the proposed multi-agent framework.

At a high level, our methodology can be divided into the following major components:

• **Data Collection:** We source our data from 18 publicly available healthcare case studies provided by the American Speech-Language-Hearing Association (ASHA). These case studies serve as authentic real-world examples of interprofessional collaboration in healthcare settings.

• **Data Preprocessing:** The unstructured case study narratives are processed into a structured JSON format suitable for LLM input. This involves both manual extraction and automated extraction using LLMs, allowing us to compare the effectiveness of both approaches.

• **Baseline Approaches:** We establish two baseline methods for treatment plan generation:

1. *Single LLM Pipeline* — The entire case study is passed to a general-purpose LLM (like GPT-4 or Claude) to generate a treatment plan.
2. *Prompt-Tuned LLM Pipeline* — The prompt is carefully crafted to simulate specific roles or expectations from the model.

• **MediSyn: Proposed Multi-Agent System:** Moving beyond single LLM baselines, we introduce MediSyn, a novel synergistic multi-agent architecture. For each case study, MediSyn dynamically identifies the healthcare team members mentioned and creates specialized agents—each representing a profession involved in the case. These agents are constrained to their domain knowledge and collaborate sequentially to generate the final treatment plan, thereby closely simulating real-world interprofessional collaboration.

• **Evaluation Metrics:** To rigorously evaluate and compare the outputs of the Single LLM, Prompt-Tuned LLM, and MediSyn systems, we employ both automated and human-centric evaluation methods:

- BLEU Score
- ROUGE
- LLM-as-a-Judge evaluations (for qualitative and contextual judgment)

3.1 Data Description

The dataset used in this study consists of a curated collection of real-world healthcare case studies sourced from the American Speech-Language-Hearing Association (ASHA) [20]. These case studies are publicly available on the ASHA IPE/IPP portal and are specifically designed to highlight examples of interprofessional collaboration across diverse healthcare scenarios.

Each case study presents a unique clinical situation involving patients from varying demographics and medical backgrounds, including conditions such as cleft palate, autism spectrum disorder, NICU interventions, and rehabilitation settings. Importantly, each case involves a multidisciplinary team working together to provide holistic patient care. Every case study in this dataset typically contains:

- A clinical background of the patient

- A list of healthcare team members involved (e.g., Nurse, Physician, Speech-Language Pathologist, Audiologist, Occupational Therapist)
- The collaborative assessment process and challenges faced
- The treatment plan co-developed by the team
- Patient outcomes and post-intervention progress

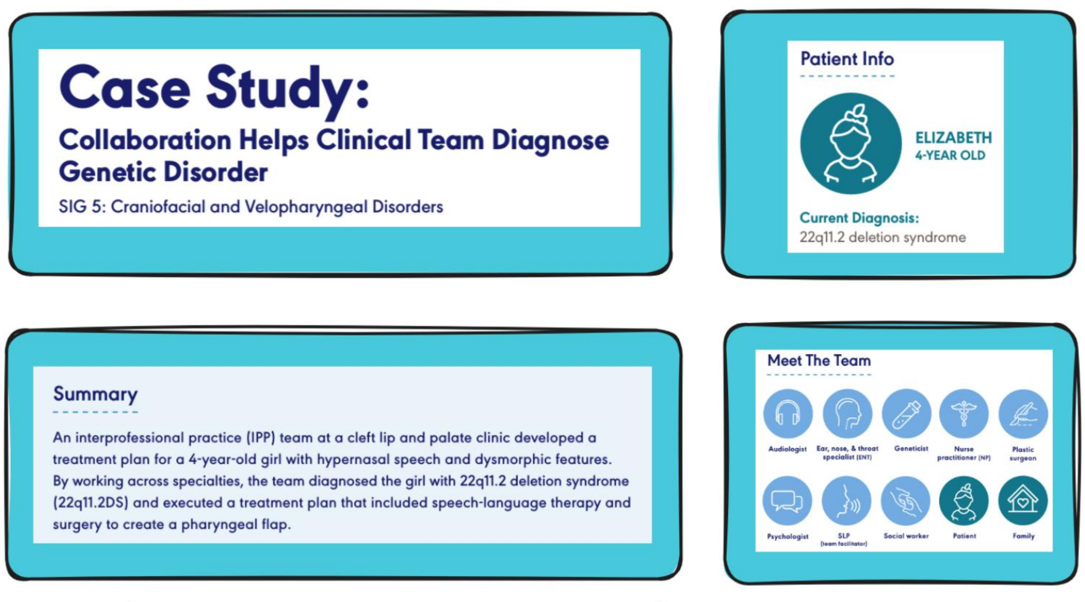


Figure 1: 1st page of the case study with Patient Info, summary, and the team members

The above diagram is a view of some of the elements in one particular case study. Each ASHA case study follows a more-or-less consistent structure comprising several distinct sections or headers, namely:

- **Summary:** A brief narrative of the case and its significance
- **Patient Info:** Demographic and clinical details about the patient
- **Meet The Team:** List of interprofessional healthcare team members involved
- **Background:** Detailed description of the patient's history, presenting problems, and symptoms
- **How They Collaborated:** Narrative explanation of the collaborative process among team members
- **Outcome:** The outcome of the case
- **Ongoing Collaboration:** Whether the team is still involved in the treatment or patient care process
- **History and Concerns:** Often a copy of the background section
- **Assessment Plan:** The roles and responsibilities of each professional in evaluating the patient
- **Assessment Results:** Key diagnostic findings from each professional's evaluation
- **IPP Treatment Plan:** The final collaboratively agreed treatment plan

- **Treatment Outcomes:** Follow-up details on patient progress
- **Team Follow-Up:** Plan for continued interprofessional communication and monitoring

To understand the features better, here's a snapshot of the raw data:

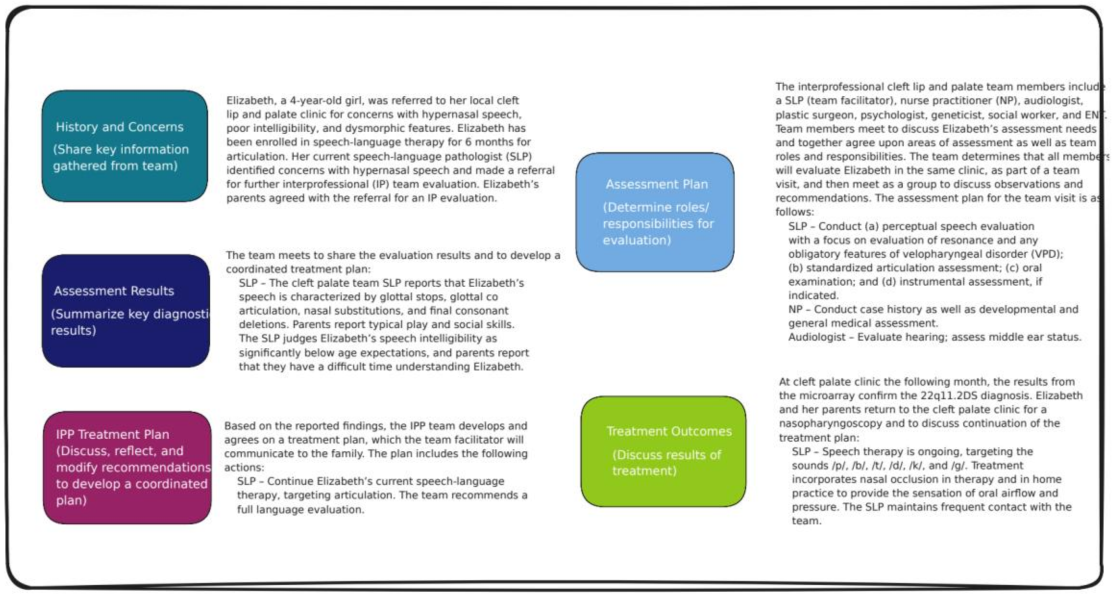


Figure 2: Data Features

In total, the dataset contains 18 such case studies. These cases serve two main purposes in our research:

1. As input data for generating treatment plans using baseline single LLM approaches.
2. As structured case scenarios for instantiating profession-specific agents within the MediSyn multi-agent framework.

Out of the 18 case studies available at the time of our work, we utilized 15 for our experiments. The remaining 3 were excluded from analysis due to missing a documented Interprofessional Practice (IPP) treatment plan and other critical components necessary for evaluation and generation. This dataset is particularly well-suited for simulating IPE scenarios as it authentically captures real-world interprofessional teamwork, role-specific responsibilities, and collaborative decision-making—essential elements for modeling and evaluating agentic behavior in our proposed system.

3.2 Data Preprocessing

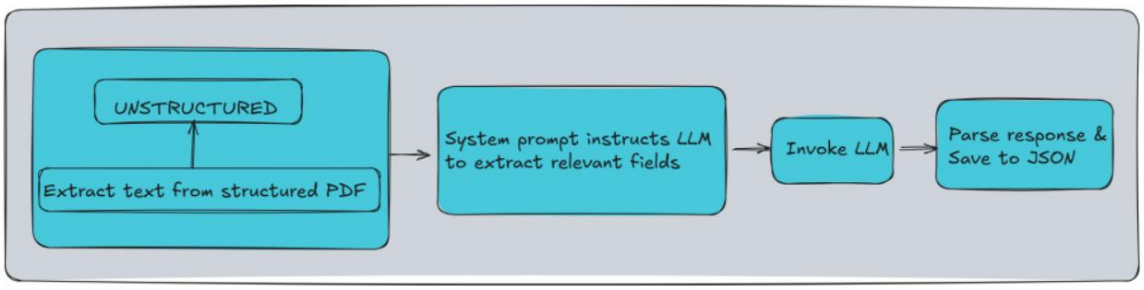


Figure 3: Data Extraction Process

Now that we have talked about the various features that a case study has. For the purpose of our work, we identified and extracted the following essential features from each case study, which directly support the downstream treatment plan generation and multi-agent modeling tasks:

Field	Description
patient_info	Includes patient's name, age, and diagnosis.
summary	Summary of the case.
team_members	List of involved professionals.
background	Patient's background.
assessment_plan	Evaluation responsibilities assigned to each professional.
assessment results	Results of diagnostic evaluations.
treatment_plan	The final Interprofessional Practice (IPP) treatment plan co-developed by the team.

The extracted information was organized into a clean JSON structure as shown below:

```
{
  "patient_info": {
    ...
  },
  "summary": "...",
  "team_members": [
    ...
  ],
  "background": "...",
  "assessment_plan": "...",
  "assessment_results": "...",
  "treatment_plan": "..."
}
```

This structured representation enables a consistent input format for all subsequent methods — including baseline LLM approaches and our proposed MediSyn multi-agent system. Thing to note is the treatment plan is our target feature. To transform the ASHA case studies into the above structured format, we employed two complementary data preprocessing strategies: (1) manual extraction, and (2) automated extraction using LLMs. These pipelines enabled consistent representation of each case study while also facilitating a comparison between human- and machine-curated inputs

3.2.1 Manual

In the manual preprocessing approach, the data from each ASHA case study was directly extracted from the raw PDF files without any alterations or modifications. Specifically, the relevant sections corresponding to the identified features were carefully copy-pasted from the case studies and structured into JSON format as described earlier. This method ensures that the extracted data maintains maximum fidelity to the original case study content, preserving the exact phrasing, structure, and information as presented in the source. The manually curated JSON files therefore serve as the most authentic representation of the data and act as the ground truth for comparison against automated methods. This approach, while labor-intensive, was crucial in providing clean, reliable data for evaluating the performance of LLM-based extraction methods and the subsequent treatment plan generation.

3.2.2 LLM

In contrast to the manual extraction approach, we also explored an automated extraction pipeline using state-of-the-art Large Language Models (LLMs) to convert the ASHA case studies from raw PDFs into a structured JSON format.

We utilized three different LLMs to perform this task:

- **LLaMA 3.2 70B Instruct model** (accessed via AWS Bedrock)
- **Claude 3.5 Sonnet** (accessed via AWS Bedrock)
- **GPT-4o** (accessed via OpenAI API)

Each of these models was provided with the raw text extracted from the PDF files using the *unstructured* Python library, along with a carefully crafted prompt instructing the model to extract all relevant information and output it in the exact JSON format described earlier. The prompt structure remained largely consistent across models, with minor adjustments specific to each model's formatting requirements. The detailed design and content of the prompt used for this extraction process is discussed in Section 4.1.

We leveraged the *unstructured* library to extract structured text from PDFs. A specialized system prompt was then created to instruct each model to extract the specific fields. This prompt was passed to the LLaMA

3.2 model deployed on AWS Bedrock via the Bedrock Runtime client. The generated response was parsed to extract valid JSON content, which was subsequently saved into a structured JSON file.

We repeated the same extraction process individually for all three models, thereby generating three separate sets of JSON files—one from each LLM. These generated datasets were then used with their respective models for treatment plan generation. Specifically:

- **LLaMA-generated JSONs** were used as input when evaluating LLaMA's performance for treatment plan generation.
- **Claude-generated JSONs** were used with Claude for treatment generation.
- **GPT-generated JSONs** were used with GPT for treatment generation.

Further details are explained in Sections 5.2, 5.3.2, and 5.4.2.

3.3 Baseline Approach

As a foundational baseline, we first explored a straightforward single-LLM approach for treatment plan generation. In this method, a general-purpose LLM is prompted with the structured case study data—in JSON format—and tasked with generating a comprehensive treatment plan for the patient described.

This approach serves two primary purposes:

1. It establishes a baseline performance level for evaluating how well a single, monolithic LLM can handle interprofessional medical reasoning.
2. It allows us to contrast the effectiveness of centralized, undifferentiated AI decision-making with our proposed multi-agent collaborative system, MediSyn.

In the single-LLM setup, we use three different models for comparative analysis:

- **GPT-4o** (OpenAI)
- **Claude 3.5 Sonnet** (Anthropic via AWS Bedrock)
- **LLaMA 3.2 70B Instruct** (Meta via AWS Bedrock)

Each model is provided the same structured input (case study JSON), which includes key components such as patient information, background, team members, assessment plan, and assessment results. The final field—the treatment plan—is left blank and is expected to be generated by the LLM itself based on the context provided in the other fields. The specific prompts used for each LLM are documented in Section 4.1.

This single-model approach provides a valuable reference point for evaluating how general-purpose LLMs perform in the context of interprofessional treatment planning in healthcare decision-making, particularly when replicating the nuances of interprofessional collaboration. As we demonstrate in our results, while these models are capable of generating plausible treatment plans, they may lack the role-specific depth and collaborative dynamics found in interprofessional teams—which our multi-agent system is designed to simulate and explore.

3.4 Prompt Tuned LLM

To improve the performance of general-purpose LLMs without altering their architecture, we explored a prompt-tuning strategy. In this approach, we continued using the same structured case study JSON as input, but paired it with a carefully engineered system prompt designed to elicit more profession-aware, collaborative treatment plans. The goal was to simulate interprofessional reasoning within a single LLM instance, without relying on an explicit multi-agent structure. Each LLM was prompted with a specific instruction that required it to produce a treatment plan, explicitly considering the perspectives of each team member involved in the case. These prompts included role-awareness cues and formatting constraints to encourage more context-sensitive outputs. The complete prompt template used in this setting is documented in Section 4.1. The use of this structured, role-aware prompt was intended to guide the model toward producing outputs that better reflected the interprofessional dynamics and specialization seen in real healthcare settings. Importantly, while the model remained a single monolithic agent, the prompt encouraged it to emulate role segmentation and simulate internal collaboration among distinct clinical perspectives. We applied this prompt-tuned approach to all three LLMs under study—GPT-4o, Claude 3.5 Sonnet, and LLaMA 3.2—using their respective prompt syntax conventions. As detailed in Section 4, the prompt-tuned variant often improved results on both automatic and human evaluation metrics compared to the baseline LLM setup, particularly in terms of plausibility, specificity, and overall contextual relevance. However, these gains varied across models and metrics, emphasizing that prompt tuning provides a meaningful but not universally optimal enhancement.

3.5 MediSyn: Proposed Multi-Agent System

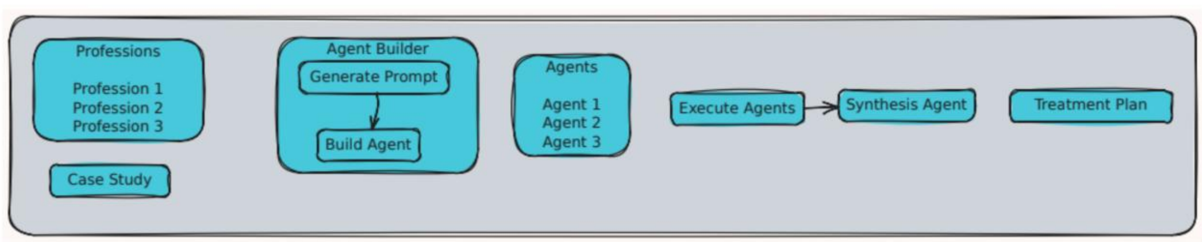


Figure 4: MediSyn Architecture

To address the limitations of monolithic and prompt-tuned LLM approaches in replicating interprofessional collaboration, we propose MediSyn, a dynamic multi-agent system (MAS) designed explicitly for IPE case studies. MediSyn simulates a virtual care team by instantiating role-specific agents aligned with the professions identified in each case study. Unlike traditional MAS architectures with static agent roles, MediSyn dynamically configures agents based on case-specific team composition, enabling flexible adaptation to diverse healthcare scenarios.

As illustrated in Figure 4, the system workflow begins with the Agent Builder, which parses the input JSON (structured from real-world IPE case studies) to extract the list of healthcare roles involved. For each role, MediSyn invokes a Prompt Generator module to construct a profession-specific prompt. These prompts encode domain knowledge, scope of practice, and context-specific responsibilities.

Next, the Agent Instantiation process launches individual agents, each constrained to reason within its domain. Each agent independently reviews the shared patient case and generates a treatment recommendation from its professional perspective. This step mirrors real-world interprofessional dynamics where specialists offer distinct but complementary contributions. Then MediSyn activates a Synthesis Agent, responsible for integrating the diverse role-specific outputs. The synthesis prompt instructs the agent to consolidate these recommendations into a coherent, role-attributed treatment plan that preserves professional distinctions. The resulting plan is presented in a standardized format with clearly labeled contributions from each team member, aligning with IPE/IPP documentation standards.

Importantly, MediSyn is model-agnostic. The system's modular architecture allows experimentation with different LLM backends—GPT-4o, Claude 3.5, or LLaMA 3.2—at both the agent and synthesis levels. This design facilitates fine-grained benchmarking of agent-level and system-level performance across models. The system serves as both a research platform and a pedagogical tool for evaluating and enhancing AI support for interprofessional collaboration in clinical settings.

3.6 Evaluation Metrics

To rigorously compare the performance of the baseline and proposed treatment generation pipelines, we employed a blend of quantitative and qualitative evaluation strategies. Specifically, we evaluated each system—Single LLM, Prompt-Tuned LLM, and MediSyn—using two established automated metrics from the field of natural language generation (BLEU and ROUGE), as well as a qualitative assessment via an "LLM-as-a-Judge" approach (resulting to quantifiable values). This multi-faceted evaluation design allows us to triangulate both linguistic fidelity and clinical plausibility across methods.

3.6.1 BLEU Score

The Bilingual Evaluation Understudy (BLEU) score is a widely adopted metric for assessing the overlap between a candidate sequence and one or more reference sequences. Originally developed for machine translation, BLEU has since found utility in various generation tasks, including clinical text generation. In our setup, we compute BLEU scores to evaluate how closely the generated treatment plans match the manually extracted reference plans. We consider unigram to 4-gram overlaps to balance both local and extended phrasal fidelity. The BLEU score is formally computed as:

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

where p_n denotes the precision of n-grams, w_n are weights (commonly uniform: $w_n = 1/N$), and BP is the brevity penalty defined as:

$$\mathrm{BP} = \begin{cases} 1 & \text{if } c > r \\ \exp(1 - \frac{r}{c}) & \text{if } c \leq r \end{cases}$$

Here, c is the length of the candidate output and r is the length of the reference.

3.6.2 ROUGE Score

The Recall-Oriented Understudy for Gisting Evaluation (ROUGE) score is another cornerstone metric for evaluating text generation, particularly useful for summarization and structured synthesis tasks. Unlike BLEU, which emphasizes precision, ROUGE prioritizes recall, making it well-suited for capturing completeness in medical plans. We focus on ROUGE-N (with N = 1, 2) and ROUGE-L (Longest Common Subsequence) in our evaluations. The ROUGE-N score is calculated as:

$$\text{ROUGE-N} = \frac{\sum_{\text{ref} \in \text{References}} \sum_{\text{gram}_n \in \text{ref}} \text{Count}_{\text{match}}(\text{gram}_n)}{\sum_{\text{ref} \in \text{References}} \sum_{\text{gram}_n \in \text{ref}} \text{Count}(\text{gram}_n)}$$

where $\text{Count}_{\text{match}}$ refers to the number of overlapping n-grams between the candidate and reference, and Count is the total number of n-grams in the reference.

3.6.3 LLM as a Judge

Beyond surface-level similarity, we incorporated a qualitative evaluation framework by prompting LLMs to act as domain-aware evaluators of generated treatment plans. Specifically, we utilized **GPT-4o** as the evaluating model due to its strong general-purpose reasoning capabilities, fluency, and high instruction-following accuracy. These characteristics make GPT-4o particularly well-suited for evaluating nuanced clinical outputs that demand context sensitivity and professional judgment.

This evaluation strategy is inspired by recent work exploring the use of LLMs as evaluators in medical generation tasks, particularly in the context of clinical correctness and plausibility [21]. It allows us to assess the outputs across multiple clinical axes—namely *plausibility*, *specificity*, *accuracy*, and *omission*.

Each plan is blindly reviewed by an LLM, which is instructed to rate it on a 1–5 scale for each of the following criteria:

- **Accuracy:** Correctness and clinical relevance of recommendations.
- **Plausibility:** Whether the recommendations are medically sound and free from hallucinations.
- **Specificity:** Degree of detail provided (e.g., “hearing screening” vs. “bilateral tympanometry screening with follow-up”).
- **Omission:** Penalization for missing crucial clinical actions or roles.

3.7 BRIDGE: Platform for IPE Education

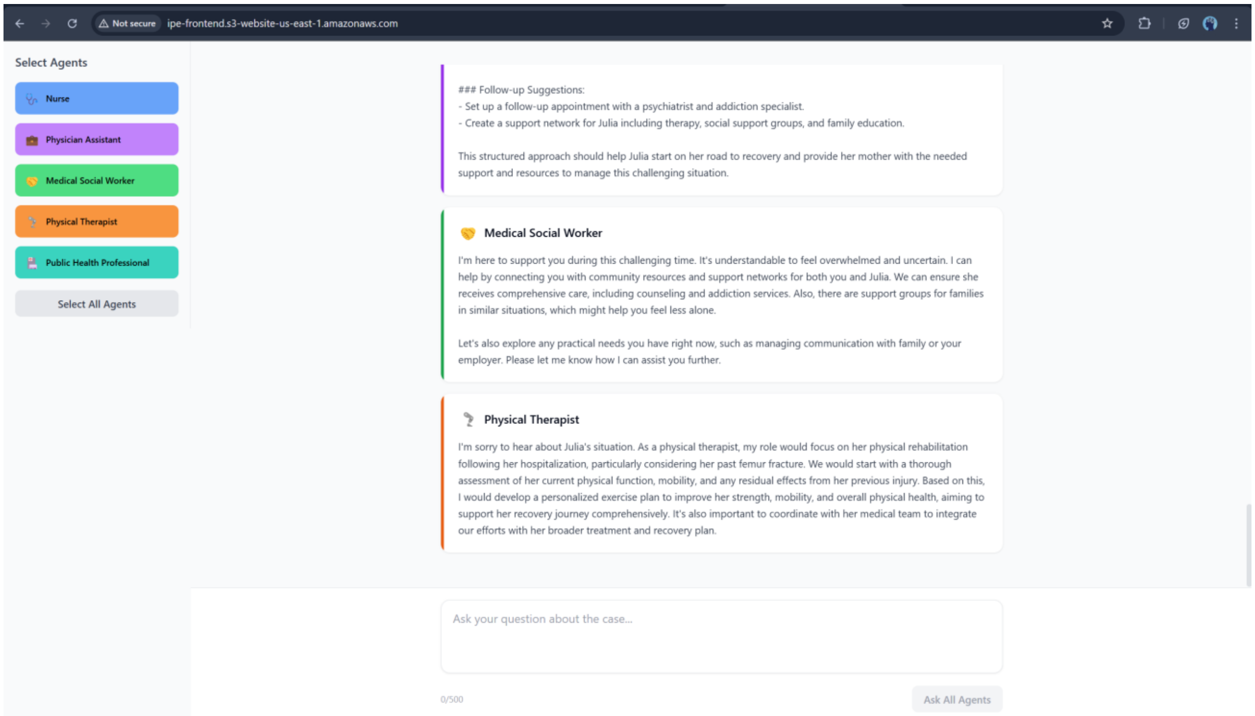


Figure 5: BRIDGE Architecture

A separate platform named "BRIDGE" was developed as a learning tool for IPE education. Bridge utilizes five pre-defined agents representing:

- Nurse
- Physician Assistant
- Medical Social Worker
- Physical Therapist
- Public Health Professional

This platform enables students to learn from interactions with these selected agents, simulating a collaborative healthcare environment.

4 Results and Discussion

To evaluate the effectiveness of different language model configurations in generating IPP treatment plans, we conducted a series of experiments comparing three approaches: a single general-purpose LLM (base), a prompt-tuned LLM, and our proposed MediSyn multi-agent system. The objective was to assess the degree to which each method could produce outputs that were accurate, plausible, specific to team member roles, and complete in terms of clinical content coverage.

All three methods were tested across three LLMs—GPT-4o, Claude 3.5 Sonnet, and LLaMA 3.2—under two data settings: one using case studies manually preprocessed into structured JSON, and the other using LLM-extracted structured data. We report results acrossa variety of metrics. Each method was applied consistently across all models, allowing for a controlled and comparative analysis.

Rather than seeking to identify a single best-performing system, our goal is to understand how each approach behaves under different evaluation conditions and to explore the strengths and trade-offs inherent in prompt tuning and agent-based reasoning strategies.

4.1 Prompt Engineering

Prompt engineering is the practice of designing and refining input instructions—known as prompts—to guide large language models (LLMs) toward producing desired outputs without altering their underlying parameters [22]. This technique leverages natural language cues to elicit specific behaviors from LLMs, enabling them to perform a wide range of tasks effectively. In healthcare, prompt engineering has been instrumental in various applications, such as enhancing patient–provider communication, supporting medical education, and facilitating personalized care and shared decision-making [23].

For instance, well-crafted prompts have been used to generate patient-friendly explanations of medical concepts, aiding in bridging the knowledge gap between clinicians and patients [24]. Additionally, prompt engineering has supported clinical decision-making by guiding LLMs to provide recommendations aligned with evidence-based guidelines [25]. These examples underscore the potential of prompt engineering to enhance the utility of LLMs in complex, domain-specific scenarios within the healthcare sector.

In our project, prompt engineering was integral to four key stages of the pipeline:

1. **Structured Data Extraction:** We developed prompts that guided LLMs to parse unstructured, narrative-style case study PDFs into a structured JSON schema.
2. **Treatment Plan Generation:** We designed prompts to instruct LLMs to produce comprehensive treatment plans by explicitly considering the perspectives of each team member involved in a case. These prompts incorporated role-awareness cues.
3. **Multi-Agent Role Synthesis:** For our proposed MediSyn architecture, we created a specialized prompt to simulate a collaborative discussion among role-specific agents. The prompt instructed the LLM to synthesize previously generated responses from multiple agents into a single unified treatment plan, capturing key contributions while resolving redundancies or contradictions.
4. **LLM-as-a-Judge Evaluation:** To complement quantitative metrics, we used LLMs as evaluators of treatment plan quality. A prompt was crafted to ask an LLM to assess and score treatment plans on multiple qualitative dimensions—such as plausibility, specificity, and completeness.

The complete prompt templates used in each of these stages are provided below.

4.1.1 Structured Data Extraction Prompts

LLaMA 3.2 Prompt to extract structured JSON from case study PDF

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
You are a medical document processor. Extract ALL information into this EXACT
    JSON format:
{
  "patient_info": {
    "name": "string",
    "age": "string",
    "diagnosis": ["string"]
  },
  "summary": "Concise clinical summary",
  "team_members": ["string"],
  "background": "Medical history and key events",
  "assessment_plan": "The assessment plan section including the individual
      team member assessments",
  "assessment_results": "The assessment results section including the
      individual team member assessments results",
  "treatment_plan": "The Treatment plan section including the individual team
      member's take on the treatment plan"
}

RULES:
1. Output ONLY valid JSON
2. Use exact field names
3. Never add comments
4. Maintain original medical terms
5. Be explanatory and don't summarize the information
<|end_header_id|>
<|start_header_id|>user<|end_header_id|>
PDF CONTENT:
{raw_pdf_content}
<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Claude 3.5 Prompt to extract structured JSON from case study PDF

```
System:
You are a medical document processor. Extract ALL information into this EXACT
     JSON format:
'''json
{
  "patient_info": {
    "name": "string",
    "age": "string",
    "diagnosis": ["string"]
  },
  "summary": "Concise clinical summary",
  "team_members": ["string"],
  "background": "Medical history and key events",
  "assessment_plan": "The assessment plan section including the individual
       team member assessments",
  "assessment_results": "The assessment results section including the
       individual team member assessments results",
  "treatment_plan": "The Treatment plan section including the individual team
       member's take on the treatment plan"
}

RULES:
1. Output ONLY valid JSON between '''json markers
2. Use exact field names
3. Never add comments
4. Maintain original medical terms
5. Be explanatory and don't summarize the information
6. Do not include any text outside the JSON structure""",
            "user": f"PDF CONTENT:\n{text_content}"
```

GPT-4o Prompt to extract structured JSON from case study PDF

```
You are a medical document processor. Extract ALL information into this EXACT
     format:
     "name": "string",
     "age": "string",
     "diagnosis": ["string"]
     "summary": "Concise clinical summary",
     "team_members": ["string"],
     "background": "Medical history and key events",
     "assessment_plan": "The assessment plan section including the individual
          team member assessments",
     "assessment_results": "The assessment results section including the
          individual team member assessments results",
     "treatment_plan": "The Treatment plan section including the individual
          team member's take on the treatment plan"

RULES:
1. Output ONLY Exact Format given
2. Use exact field names
3. Never add comments
4. Maintain original medical terms
5. Be explanatory and don't summarize the information
```

4.1.2 Treatment Plan Generation Prompts

Base Prompt

```
Generate a treatment plan.
```

Prompt-Tuned LLM Prompt

```
**Required Output Format**
Create a comprehensive treatment plan with specific recommendations from each
    team member's perspective.

Structure your response using EXACTLY these section headers:
{team_member_roles}
{For each team member}
[Full Team Member Role Name]: [Recommendations]
```

4.1.3 MediSyn Prompts

Prompt Generator Prompt

```
Generate a prompt for this role as a {}
```

Synthesis Agent Prompt

```
As the synthesis agent, your task is to carefully analyze and integrate the
     recommendations provided by each role-specific healthcare agent involved
     in the current clinical case. Each agent's input represents their
     specialized professional perspective.

Your goal is to synthesize these diverse insights into a cohesive, context-
     sensitive, and comprehensive treatment plan.

Follow these guidelines strictly:

1. Read all the individual recommendations thoroughly to understand each
     professional perspective.
2. Identify areas of agreement, differences, or potential conflicts among the
     recommendations.
3. Ensure that the final synthesized treatment plan respects the contributions
      from each professional perspective.
4. Clearly outline specific, actionable recommendations under each team member
     's role.

Use EXACTLY the following format for your final synthesized response:

Required Output Format
Create a comprehensive treatment plan with specific recommendations from each
     team member's perspective.

Structure your response using EXACTLY these section headers:
{team_member_roles}
```

4.1.4 Evaluation Prompt

LLM-as-a-Judge Evaluation Prompt

```
Evaluate on the basis of the following criteria:
The diagnostic accuracy framework was categorized into the following
     components:
(1) Overall Accuracy
(2) Plausibility
(3) Specificity
(4) Omission / Uncertainty

Accuracy represents how well it meets the definition of a diagnosis.
Plausibility evaluates whether the diagnosis is reasonable and free from
     hallucinations.
Specificity captures the level of detail (e.g., sepsis vs sepsis from
     influenza pneumonia).
Omission penalizes missing clinically important diagnoses.
Uncertainty (conditional on omission) penalizes failure to use available
     information.

Score each axis independently and provide a brief justification for each.
```

Figure 6

4.2 Data Preparation Comparison - OpenAI vs Claude vs Llama

To evaluate how well different LLMs perform at extracting structured information from unstructured medical case study PDFs, we compared outputs generated by three models—GPT-4o (OpenAI), Claude 3.5 Sonnet (Anthropic), and LLaMA 3.2 70B (Meta). Each model was tasked with converting a single narrative-style ASHA case study into a predefined JSON schema that included all relevant fields, such as patient info, summary, team members, background, assessment plan, assessment results, and treatment plan.

Outputs were qualitatively evaluated through visual inspection. Rather than assigning quantitative scores, we focused on understanding whether each model captured the essential clinical information with enough fidelity to support downstream tasks like treatment plan generation. For example, even if a model paraphrased or shortened the original text, as long as it retained the key clinical facts, we considered the output usable.

To make the differences more interpretable, we provide visual comparisons for a selected subset of fields, omitting longer fields like assessment plan and assessment results for brevity. We will walk through one representative case study to illustrate our findings, noting that similar patterns were observed across 16 the other case studies as well. For brevity, we present the comparisons for patient info, summary, and treatment plan in the main report.

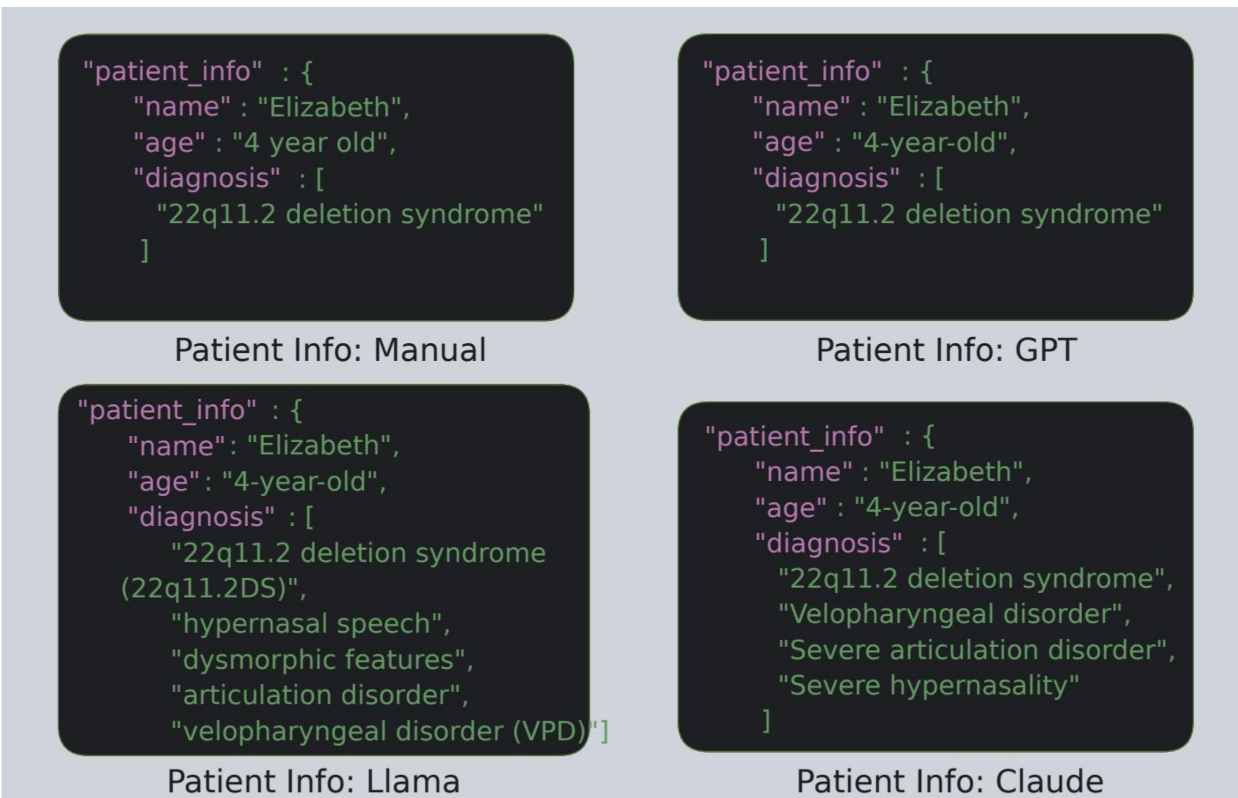


Figure 7: Comparison of patient info field

Figure 8: Team members field comparison

The patient info field contains basic demographic and diagnostic details such as the patient's name, age, and diagnosis. All three models successfully identified and extracted the patient's name as "Elizabeth" and represented the age as "4-yearold," although Claude formatted it as "4 year old" (without hyphenation), and GPT preserved the format closest to the manual version.

The most noticeable difference appeared in the diagnosis field. The manually curated JSON included only one diagnosis: "22q11.2 deletion syndrome". GPT-4o reproduced this faithfully, whereas Claude and LLaMA included additional inferred conditions such as "severe hypernasality," "velopharyngeal disorder," and "articulation disorder." While these additions are medically plausible and contextually relevant, they are not explicitly mentioned in the original text, making them mild hallucinations.

Overall, GPT-4o most closely matched the manual version in both structure and content. Claude and LLaMA tended to be more verbose or inferential in their outputs, which may reflect an attempt to enrich the response but could introduce unintended interpretations if used in a strictly extractive setting.

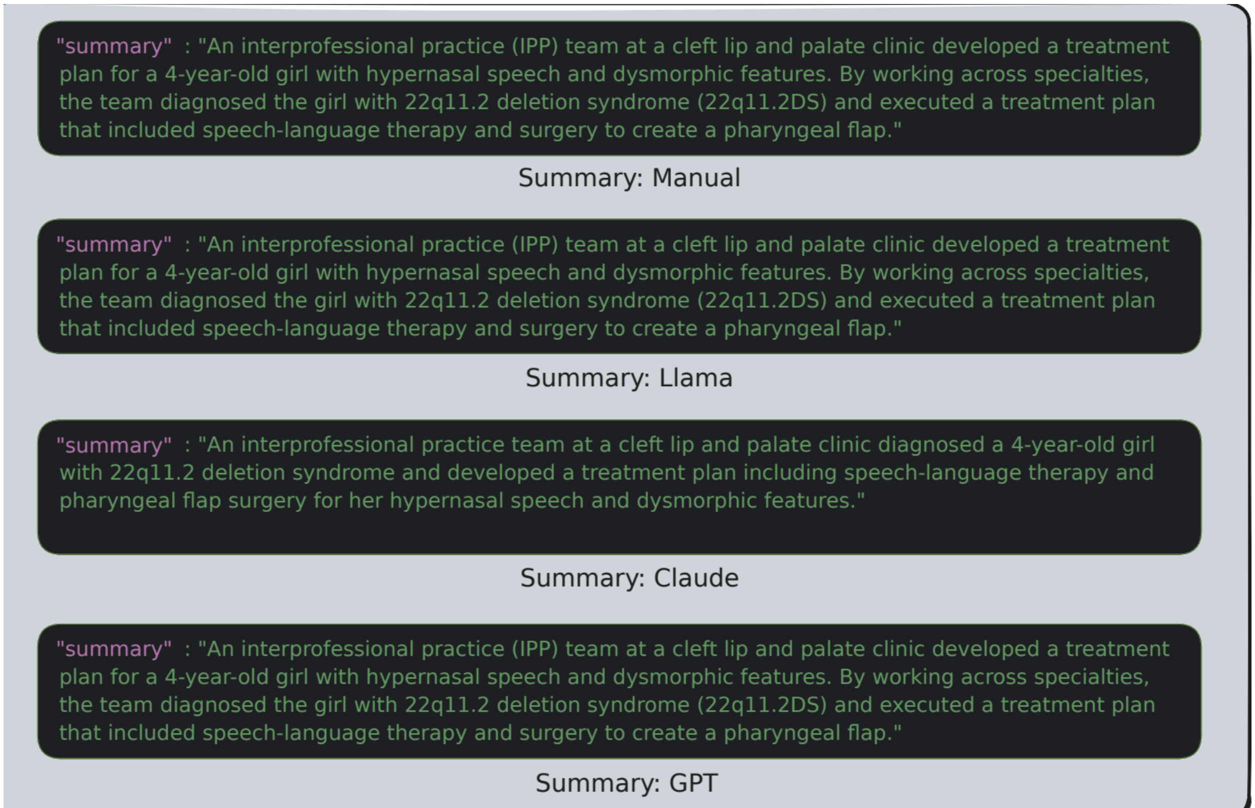


Figure 9: Comparison of summary field across manual, GPT-4o, Claude 3.5,

Both the manually extracted summary and the outputs from GPT-4o and LLaMA aligned closely in structure and wording. Claude 3.5, however, restructured the sentence slightly, opting for a more natural and narrative-driven style. It preserved the core content but combined diagnosis and planned evaluation into a smoother, paraphrased form. While the meaning remained intact, the output lacked the more factual tone and field-oriented structure seen in the manual version.

This contrast highlights a key difference in model behavior: while GPT-4o and LLaMA tend to extract text conservatively, Claude occasionally leans toward stylistic fluency, which could be beneficial for summarization tasks but may diverge from strict extraction goals.

The manually prepared version of treatment plan offers a clean breakdown with clearly delineated contributions from each role. GPT-4o generated a concise and reasonably structured plan, correctly identifying most team members and assigning plausible actions. However, its output was relatively short and omitted several role-specific recommendations found in the manual version. LLaMA and Claude 3.5 produced significantly longer and more detailed outputs. Both captured additional tasks, such as “long-term hearing monitoring” or “referrals for social support,” which were not explicitly stated in the case study but are contextually relevant. While this can add richness, it also introduces interpretive variability and potential hallucinations. Notably, both LLaMA and Claude reformatted the original structure into more prose-driven responses rather than following the explicit team-member formatting in the manual. This divergence highlights the models’ tendency to favor generative fluency over strict structural alignment when not explicitly instructed to follow a rigid schema.

Figure 10: Comparison of treatment plan field across manual, GPT-4o, Claude 3.5,

Across the five fields examined—patient info, summary, team members, background, and treatment plan—all three LLMs demonstrated the ability to extract structured information from unstructured clinical narratives with varying levels of fidelity. Although some outputs included minor hallucinations or omitted less obvious roles, the core medical context was retained in all cases.

Ultimately, the qualitative differences observed across models were not significant enough to impact downstream performance in treatment plan generation. This supports our decision to use the LLM-specific extracted JSONs as valid input for subsequent evaluation stages, without the need for strict quantitative scoring or error correction.

4.3 Treatment Plans Comparison - OpenAI vs Claude vs Llama

In this section, we qualitatively compare the final treatment plans generated by each language model under three different configurations: base LLM, prompt-tuned LLM, and our MediSyn multi-agent system. For each of the three models—LLaMA 3.2, Claude 3.5 Sonnet, and GPT-4o—we present side-by-side outputs in image form and analyze the differences in structure, completeness, and role-specificity.

This comparison is conducted using the same case study used in Section 4.2, allowing us to directly observe how changes in prompting and agentic structure influence the final treatment plan. Our analysis is based on visual inspection of the generated texts and focuses on the qualitative aspects of interprofessional reasoning, such as clarity,LLM vs. Prompt-Tuned LLM vs. MediSyn

Figure 11:Combined MediSyn treatment plans generated by LLaMA 3.2, Claude 3.5,

detail, and distribution of content across roles. The goal is to assess whether the generated plans reflect the values of IPE/IPP and preserve the fidelity of domain-specific contributions.

For reference, the manually extracted treatment plan from the original case study is provided in Figure 7, as discussed in Section 4.2. This serves as a ground truth against which the generated outputs can be informally compared.

We begin with results from LLaMA 3.2, followed by Claude 3.5 Sonnet and GPT4o.

4.3.1 Manually prepared data

The LLama MediSyn output presents the most professional and polished plan. It organizes content into discipline-based sections (e.g., Speech-Language Therapy, Genetic Counseling) using a hierarchical structure with goals and interventions. While it does not explicitly name individual agents or team members, the sections correspond to their domains, preserving the spirit of role-aware collaboration. This structure mirrors actual clinical documentation practices and offers the best balance between readability, depth, and IPE relevance

The Claude MediSyn output closely mirrors the prompt-tuned version in content and format but appears even more cohesive. The structure is preserved nearly identically, with slight improvements in transitions and alignment across roles. This version demonstrates that Claude 3.5 handles agent-based role synthesis very effectively, integrating diverse agent outputs into a unified narrative. The result reads like a polished interprofessional plan, showing that the MediSyn architecture complements Claude's strengths in clarity and formatting.

The MediSyn output from GPT-4o is well-structured, detailed, and clearly aligned with IPE principles. Each recommendation is grouped under team roles, and the language reflects collaborative planning and practical interventions. Compared to the prompt-tuned version, the MediSyn output appears more polished, both in tone and formatting. The addition of "Overall Coordination and Follow-Up" as a final section is a valuable touch, signaling awareness of care continuity. This output demonstrates that GPT-4o benefits from the MediSyn framework, producing one of its most comprehensive and context-sensitive plans.

"generated_text" : "**Treatment Plan for Elizabeth**\n\n**Short-term Goals (0-6 months)**\n\n1. **Speech-Language Therapy:**\n\t* Implement individualized speech therapy sessions (2-3 times a week) to target articulation, resonance, and intelligibility.\n\t* Focus on reducing glottal stops, glottal co-articulation, nasal substitutions, and final consonant deletions.\n\t* Develop a home program for parents to practice with Elizabeth.\n2. **Surgical Intervention:**\n\t* Schedule a pharyngeal flap surgery to address velopharyngeal dysfunction (VPD) and hypernasality.\n\t* Collaborate with the surgeon and SLP to determine the optimal timing for surgery.\n3. **Genetic Testing and Counseling:**\n\t* Complete genetic testing (microarray) to confirm the diagnosis of 22q11.2 deletion syndrome (22q11.2DS).\n\t* Provide genetic counseling to Elizabeth's parents, explaining the implications of the diagnosis on her medical care and development.\n4. **ENT Follow-up:**\n\t* Schedule regular follow-up appointments with the ENT to monitor middle ear function and hearing.\n\n**Long-term Goals (6-12 months)**\n\n1. **Speech-Language Therapy:**\n\t* Continue individualized speech therapy sessions to maintain progress and address emerging speech and language skills.\n\t* Expand therapy goals to include language development and literacy skills.\n2. **Surgical Follow-up:**\n\t* Schedule post-operative follow-up appointments with the surgeon to monitor the effectiveness of the pharyngeal flap surgery.\n\t* Collaborate with the SLP to assess the impact of surgery on Elizabeth's speech and resonance.\n3. **Multidisciplinary Team Meetings:**\n\t* Schedule regular multidisciplinary team meetings (every 3-4 months) to review Elizabeth's progress, address any concerns, and adjust the treatment plan as needed.\n4. **Family Support and Education:**\n\t* Provide ongoing education and support to Elizabeth's parents on managing her condition, including strategies for promoting speech and language development.\n\t* Facilitate connections with local support groups and resources for families affected by 22q11.2DS.\n\n**Interventions and Strategies:**\n\n1. **Speech Therapy:**\n\t* Use evidence-based speech therapy approaches, such as articulation therapy and resonance therapy.\n\t* Incorporate play-based and child-centered activities to promote engagement and motivation.\n2. **Surgical Intervention:**\n\t* Collaborate with the surgeon to determine the optimal surgical approach for Elizabeth's specific needs.\n\t* Ensure that Elizabeth's parents are well-informed and prepared for the surgical procedure.\n3. **Genetic Counseling:**\n\t* Provide emotional support and counseling to Elizabeth's parents, addressing their concerns and questions about the diagnosis.\n\t* Offer resources and referrals for additional support and information.\n4. **ENT Care:**\n\t* Collaborate with the ENT to monitor Elizabeth's middle ear function and hearing.\n\t* Develop a plan for managing ear infections and promoting healthy ear function.\n\n**Outcome Measures:**\n\n1. **Speech and Language Skills:**\n\t* Monitor progress in articulation, resonance, and intelligibility using standardized assessments and rating scales.\n\t* Track language development and literacy skills using age-appropriate assessments.\n2. **Surgical Outcomes:**\n\t* Evaluate the effectiveness of the pharyngeal flap surgery in reducing hypernasality and improving speech resonance.\n\t* Monitor post-operative complications and adjust the treatment plan as needed.\n3. **Quality of Life:**\n\t* Assess Elizabeth's overall quality of life, including her ability to communicate effectively and participate in daily activities.\n\t* Evaluate the impact of the treatment plan on Elizabeth's self-esteem and confidence.\n\nBy working together as an interprofessional team, we can provide comprehensive care to Elizabeth, addressing her complex needs and promoting optimal outcomes in speech, language, and overall development."

Llama Base: Treatment Plan (Manual)

Figure 12: LLaMA 3.2

"generated_text" : "Here is a comprehensive treatment plan with specific recommendations from each team member's perspective:\n\n**Social Worker**: Provide emotional support to Elizabeth and her family, and connect them with local resources and support groups for children with 22q11.2 deletion syndrome. Ensure that the family has access to necessary resources and services, and help them navigate the healthcare system.\n\n**Plastic Surgeon**: Perform surgical intervention to create a pharyngeal flap to address Elizabeth's velopharyngeal disorder (VPD) and hypernasal speech. Collaborate with the SLP to assess the effectiveness of the surgery and make any necessary adjustments.\n\n**Geneticist**: Confirm the diagnosis of 22q11.2 deletion syndrome through genetic testing and provide counseling to Elizabeth's family on the implications of this diagnosis for her medical care and future development. Monitor Elizabeth for any potential complications associated with this syndrome.\n\n**Audiologist**: Continue to monitor Elizabeth's hearing and middle ear function, and provide recommendations for hearing conservation and protection. Ensure that Elizabeth's hearing is optimized to support her speech and language development.\n\n**Ear, Nose, & Throat Specialist (ENT)**: Continue to monitor Elizabeth's ear health and address any recurring ear infections. Collaborate with the surgeon to assess the impact of the pharyngeal flap surgery on Elizabeth's ear health.\n\n**Nurse Practitioner (NP)**: Provide ongoing developmental and medical assessments to monitor Elizabeth's overall health and development. Collaborate with the SLP to address any speech and language delays, and provide support and guidance to Elizabeth's family.\n\n**SLP (Team Facilitator)**: Develop a comprehensive speech-language treatment plan to address Elizabeth's articulation disorder, hypernasality, and language delays. Provide regular speech therapy sessions to target these areas, and collaborate with the surgeon to assess the effectiveness of the pharyngeal flap surgery on Elizabeth's speech.\n\n**Psychologist**: Provide ongoing assessments of Elizabeth's cognitive and behavioral development, and offer support and guidance to her family on managing any behavioral challenges. Collaborate with the SLP to address any language delays and provide strategies for promoting social skills and play.\n\n**Patient**: Elizabeth will participate in regular speech therapy sessions and attend follow-up appointments with the IPP team to monitor her progress. She will also undergo surgical intervention to create a pharyngeal flap to address her VPD and hypernasal speech.\n\n**Family**: Elizabeth's family will provide ongoing support and encouragement to Elizabeth, and participate in speech therapy sessions and follow-up appointments with the IPP team. They will also seek out additional resources and support groups for children with 22q11.2 deletion syndrome."

Llama prompt-tuned: Treatment Plan (Manual)

Figure 13: : LLaMA 3.2

"generated_text" : "Based on the comprehensive assessment results provided by the interprofessional practice (IPP) team, I'll generate a treatment plan for Elizabeth, the 4-year-old patient with suspected 22q11.2 deletion syndrome (22q11.2DS). The plan will address her various needs and involve multiple specialists.\n\nTreatment Plan for Elizabeth:\n\n1. Genetic Confirmation and Counseling:\n - Await results of the microarray genetic test\n - Schedule follow-up with geneticist to discuss results and implications\n - Provide genetic counseling to family if 22q11.2DS is confirmed\n\n2. Speech and Language Therapy:\n - Implement intensive speech therapy sessions (2-3 times per week)\n - Focus on improving articulation, reducing hypernasality, and enhancing overall speech intelligibility\n - Conduct formal language assessment and incorporate language therapy if needed\n - Utilize specialized techniques for velopharyngeal dysfunction (VPD)\n\n3. Surgical Intervention:\n - Schedule velopharyngeal imaging to assess the extent of VPD\n - Based on imaging results, plan for potential pharyngeal flap surgery to address hypernasality\n - Coordinate pre- and post-surgical care with the plastic surgeon and ENT\n\n4. Audiological Monitoring:\n - Schedule regular hearing checks every 6 months\n - Monitor for potential ear infections and middle ear status\n\n5. Developmental and Psychological Support:\n - Conduct a comprehensive developmental assessment\n - Provide behavioral therapy if needed\n - Offer family counseling to help parents understand and manage Elizabeth's condition\n\n6. Medical Management:\n - Establish a primary care plan with the nurse practitioner\n - Schedule regular check-ups to monitor overall health and development\n - Create a care plan for managing potential 22q11.2DS-related issues (if confirmed)\n\n7. ENT Follow-up:\n - Monitor for recurrent ear infections\n - Assess airway regularly, watching for signs of sleep apnea\n\n8. Social Support:\n - Connect family with support groups for 22q11.2DS or similar conditions\n - Provide resources for educational support and early intervention services\n\n9. Interdisciplinary Coordination:\n - Schedule regular team meetings to discuss Elizabeth's progress\n - Adjust treatment plan as needed based on her response and any new findings\n\n10. Parent Education and Involvement:\n - Provide education on 22q11.2DS and its implications\n - Train parents in at-home speech exercises and strategies to support Elizabeth's development\n\n11. Long-term Planning:\n - Develop a long-term care plan, considering potential future needs related to 22q11.2DS\n - Plan for transition to school, including potential need for an Individualized Education Program (IEP)\n\nThis treatment plan should be reviewed and adjusted regularly based on Elizabeth's progress and any new information that becomes available, especially after confirming the genetic diagnosis. The IPP team should maintain open communication channels to ensure coordinated care across all specialties involved in Elizabeth's treatment."

Claude Base: Treatment Plan (Manual)

Figure 14: Claude 3.5,

"generated_text" : "Based on the provided case study, here is a comprehensive treatment plan with specific recommendations from each team member's perspective:\n\nSocial worker: \n- Provide resources and support for the family to navigate the healthcare system\n- Assist with insurance coverage and financial planning for ongoing treatments\n- Connect the family with support groups for 22q11.2 deletion syndrome\n- Offer counseling services to help the family cope with the diagnosis\n\nPlastic surgeon:\n- Schedule and perform pharyngeal flap surgery to address velopharyngeal insufficiency\n- Conduct post-operative follow-ups to ensure proper healing and function\n- Collaborate with the SLP to monitor speech improvements after surgery\n\nGeneticist:\n- Confirm 22q11.2 deletion syndrome diagnosis through genetic testing\n- Provide genetic counseling to the family, explaining the implications of the diagnosis\n- Recommend screening for other potential medical issues associated with 22q11.2DS\n- Schedule regular follow-ups to monitor for any emerging symptoms related to the syndrome\n\nAudiologist:\n- Conduct regular hearing assessments to monitor for any changes\n- Provide education on the importance of hearing protection\n- Monitor middle ear function, given the history of ear infections\n\nEar, nose, & throat specialist (ENT):\n- Monitor for recurrent ear infections and provide treatment as needed\n- Assess airway function regularly, watching for any signs of sleep apnea\n- Collaborate with the plastic surgeon on pre- and post-operative care\n\nNurse practitioner (NP):\n- Coordinate care between different specialists\n- Monitor Elizabeth's overall health and development\n- Provide education to the family on 22q11.2DS and its management\n- Assist in medication management if needed\n\nSLP (team facilitator):\n- Continue speech-language therapy, focusing on articulation and resonance\n- Conduct formal language assessment and provide therapy as needed\n- Collaborate with the plastic surgeon on pre- and post-operative speech assessments\n- Provide regular updates to the team on Elizabeth's speech and language progress\n\nPsychologist:\n- Conduct a comprehensive developmental and behavioral assessment\n- Provide support for any emotional or behavioral challenges that may arise\n- Monitor cognitive development and recommend interventions if needed\n- Offer family counseling to help cope with the diagnosis and treatment process\n\nPatient:\n- Participate in speech therapy sessions and practice exercises at home\n- Undergo necessary medical procedures, including pharyngeal flap surgery\n- Attend regular check-ups with various specialists as scheduled\n\nFamily:\n- Educate themselves about 22q11.2 deletion syndrome and its implications\n- Support Elizabeth in her therapy and medical treatments\n- Attend genetic counseling sessions to understand the diagnosis fully\n- Implement strategies at home to support Elizabeth's speech and language development\n- Participate in support groups for families affected by 22q11.2DS\n- Maintain open communication with the healthcare team about Elizabeth's progress and any concerns"

Claude prompt-tuned: Treatment Plan (Manual)

Figure 15: Claude 3.5

"generated_text" : "### Treatment Plan for Elizabeth\n\n**1. Speech-Language Therapy**\n - **Frequency**: 2 sessions per week\n - **Goals**:\n - Address severe articulation disorder: Focus on reducing glottal stops, nasal substitutions, and final consonant deletions.\n - Improve speech intelligibility and resonance.\n - **Methods**:\n - Articulation exercises targeting specific phonemes.\n - Resonance therapy using visual and auditory feedback.\n - Incorporate play-based activities to enhance engagement.\n\n**2. Surgical Intervention**\n - **Procedure**: Pharyngeal flap surgery\n - **Objective**: Correct velopharyngeal insufficiency to reduce hypernasality.\n - **Timeline**: Schedule post confirmation of genetic testing.\n\n**3. Genetic Counseling and Testing**\n - **Objective**: Confirm diagnosis of 22q11.2 deletion syndrome (22q11.2DS).\n - **Follow-up**: Discuss results and any additional medical implications with Elizabeth’s parents.\n\n**4. Additional Language Assessment**\n - **Purpose**: Evaluate specific language delay, as indicated by screening failure.\n - **Action**: Conduct comprehensive formal language assessment.\n\n**5. Monitoring and Management by Healthcare Professionals**\n - **ENT**: Regular monitoring of middle ear function, especially due to history of ear infections.\n - **Nurse Practitioner**: Ongoing developmental and general health assessments.\n\n**6. Behavioral and Psychological Support**\n - **Psychologist**: Evaluate development and behavior, providing support to parents to manage any behavioral concerns linked to 22q11.2DS.\n\n**7. Family and Community Support**\n - **Social Worker**: \n - Identify any potential barriers to care and assist the family in accessing resources.\n - Connect family with support groups specific to 22q11.2DS.\n\n**8. Audiology Follow-Up**\n - **Objective**: Regular audiological assessments to monitor hearing.\n\n**Coordination and Review**\n- **IPP Team Meetings**: Monthly meetings to review progress, coordinate care, and update the treatment plan as necessary.\n- **Parent Involvement**: Engage Elizabeth’s parents in treatment decisions and provide education on 22q11.2DS and its potential impacts.\n\nThis comprehensive treatment plan aims to address Elizabeth’s speech and language needs, confirm the diagnosis of 22q11.2DS, and provide support across multiple facets of her medical and developmental care."

GPT Base: Treatment Plan (Manual)

Figure 16: GPT-4o

"generated_text" : "['Social worker', 'Plastic surgeon', 'Geneticist', 'Audiologist', 'Ear, nose, & throat specialist (ENT)', 'Nurse practitioner (NP)', 'SLP (team facilitator)', 'Psychologist', 'Patient', 'Family']\n\n**Social worker**: Discuss potential barriers such as financial constraints or access to specialized care. Provide resources for support groups or community programs that can aid the family in managing Elizabeth’s condition. Encourage connecting with other families experiencing similar challenges for mutual support.\n\n**Plastic surgeon**: Work with the SLP on planning the timing and type of surgery for creating a pharyngeal flap to address the velopharyngeal disorder and hypernasality. Collaborate on VP imaging to ensure comprehensive pre-surgical assessment.\n\n**Geneticist**: Recommend follow-up after receiving microarray results to confirm the diagnosis of 22q11.2 deletion syndrome. Explain the broader implications of the condition, such as cardiac monitoring or immunological assessment. Liaise with other specialists to plan long-term care.\n\n**Audiologist**: Conduct regular hearing evaluations to monitor for any changes in hearing status, particularly given Elizabeth’s history of ear infections. Provide recommendations for strategies to maintain healthy ear conditions and prevent future infections.\n\n**Ear, nose, & throat specialist (ENT)**: Monitor Elizabeth's ear health given the history of ear infections. Ensure normal middle ear function is maintained. Continue periodic evaluations for potential issues related to the airway in context with her 22q11.2DS diagnosis.\n\n**Nurse practitioner (NP)**: Coordinate ongoing general medical assessment and developmental surveillance. Address any emerging health concerns related to 22q11.2DS, ensuring timely referrals to appropriate specialists. Provide education to Elizabeth’s family about managing her condition.\n\n**SLP (team facilitator)**: Implement a tailored speech therapy program focusing on articulation, compensatory strategies for hypernasality, and language development. Facilitate collaboration across the IPP team to coordinate interventions and follow-ups. Ensure parents are engaged in home practice activities.\n\n**Psychologist**: Offer developmental support and behavioral strategies to help Elizabeth in social and educational settings. Monitor her psychological well-being over time and provide interventions as necessary. Support the family in understanding Elizabeth’s development in the context of 22q11.2DS.\n\n**Patient**: Elizabeth should continue to participate in speech-language therapy regularly and attend all recommended medical appointments. Her engagement in play and social interaction should be encouraged to support her social skills development.\n\n**Family**: Prioritize attendance at all healthcare appointments, and actively participate in Elizabeth’s therapy sessions and home practices. Seek educational resources about 22q11.2DS to better understand and support Elizabeth’s unique needs. Maintain communication with the IPP team for coordinated care and support."

GPT prompt-tuned: Treatment Plan (Manual)

Figure 17: GPT-4o

```
"generated_text" : "**Treatment Plan for Elizabeth**\n\n**Short-Term Goals (0-6 months)**\n\n1. **Speech-Language Therapy:**\n\t* Focus on improving articulation and reducing hypernasal speech through targeted exercises and strategies.\n\t* Implement individualized speech therapy sessions (2 times a week) to address glottal stops, glottal co-articulation, nasal substitutions, and final consonant deletions.\n\t* Educate parents on home-based exercises and activities to support speech development.\n2. **Surgical Intervention:**\n\t* Schedule surgery to create a pharyngeal flap to improve velopharyngeal function and reduce hypernasal speech.\n\t* Coordinate with the surgeon to ensure timely and effective surgical intervention.\n3. **Audiological Monitoring:**\n\t* Schedule regular hearing tests (every 3 months) to monitor middle ear function and hearing status.\n\t* Provide ear infection prevention strategies to parents to minimize the risk of future infections.\n4. **Genetic Counseling:**\n\t* Provide genetic counseling to Elizabeth's parents to educate them on 22q11.2DS, its implications, and potential long-term effects.\n\t* Offer support and resources for managing the condition.\n5. **Interdisciplinary Team Meetings:**\n\t* Schedule regular team meetings (every 2 months) to discuss Elizabeth's progress, address concerns, and adjust the treatment plan as needed.\n\n**Long-Term Goals (6-12 months)**\n\n1. **Speech-Language Therapy:**\n\t* Continue individualized speech therapy sessions to refine articulation skills and improve overall communication.\n\t* Gradually increase speech therapy sessions to 3 times a week as needed.\n2. **Surgical Follow-up:**\n\t* Schedule post-surgical follow-up appointments with the surgeon to monitor the effectiveness of the pharyngeal flap and address any complications.\n3. **Developmental Monitoring:**\n\t* Collaborate with the psychologist to monitor Elizabeth's developmental progress and address any concerns or delays.\n4. **Family Support:**\n\t* Provide ongoing support and resources to Elizabeth's family to help them cope with the diagnosis and manage the condition.\n5. **Community Referrals:**\n\t* Identify and refer Elizabeth to community-based services, such as early childhood education programs, to support her overall development.\n\n**Roles and Responsibilities:**\n\n1. **Speech-Language Pathologist (SLP):** Provide individualized speech therapy, educate parents on home-based exercises, and monitor speech progress.\n2. **Surgeon:** Perform pharyngeal flap surgery, monitor post-surgical progress, and address any complications.\n3. **Audiologist:** Conduct regular hearing tests, monitor middle ear function, and provide ear infection prevention strategies.\n4. **Geneticist:** Provide genetic counseling, educate parents on 22q11.2DS, and offer support and resources.\n5. **Psychologist:** Monitor developmental progress, address concerns or delays, and provide support to Elizabeth and her family.\n6. **Social Worker:** Identify and provide community resources, address potential barriers to care, and offer support to Elizabeth's family.\n\n**Evaluation and Revision:**\n\nThe treatment plan will be regularly evaluated and revised as necessary based on Elizabeth's progress, feedback from the interdisciplinary team, and input from Elizabeth's family."
```


Llama Base: Treatment Plan (LLM)

Figure 18: LLaMA 3.2:


```
"generated_text" : "Here is the comprehensive treatment plan with specific recommendations from each team member's perspective:\n\n**Audiologist**: Recommend continued monitoring of Elizabeth's hearing and middle ear function, especially given her history of ear infections. Provide education to parents on the importance of regular hearing screenings and ear infection management.\n\n**Ear, nose, & throat specialist (ENT)**: Recommend continued monitoring of Elizabeth's ear infections and middle ear function. Consider tympanometry to assess middle ear mobility and function. Provide guidance on ear infection prevention and management strategies.\n\n**Geneticist**: Confirm the diagnosis of 22q11.2 deletion syndrome (22q11.2DS) through genetic testing and provide genetic counseling to Elizabeth's parents. Educate them on the implications of this diagnosis, including potential developmental and medical sequelae. Offer referrals to other specialists as needed (e.g., cardiologist, immunologist).\n\n**Nurse practitioner (NP)**: Continue to monitor Elizabeth's overall health and development, focusing on communication skills and potential developmental delays. Provide education to parents on strategies to support Elizabeth's communication development. Refer to other specialists (e.g., occupational therapist, physical therapist) as needed.\n\n**Plastic surgeon**: Recommend surgical intervention to create a pharyngeal flap to address Elizabeth's velopharyngeal disorder (VPD) and hypernasal speech. Collaborate with the SLP to ensure optimal timing and coordination of surgery with speech therapy.\n\n**Psychologist**: Assess Elizabeth's developmental and behavioral needs, including potential cognitive and emotional implications of 22q11.2DS. Provide recommendations for behavioral interventions and referrals to other specialists (e.g., child psychologist, developmental pediatrician) as needed.\n\n**SLP (team facilitator)**: Develop a comprehensive speech-language treatment plan to address Elizabeth's articulation disorder, hypernasal speech, and velopharyngeal disorder (VPD). Collaborate with the plastic surgeon to ensure optimal coordination of speech therapy with surgical intervention. Provide education to parents on strategies to support Elizabeth's communication development.\n\n**Social worker**: Identify potential barriers to care (e.g., financial, transportation, childcare) and provide resources and referrals to address these barriers. Offer emotional support and counseling to Elizabeth's parents, as needed. Connect them with local support groups and resources for families affected by 22q11.2DS."
```


Llama prompt-tuned: Treatment Plan (LLM)

Figure 19: LLaMA 3.2:


```
"generated_text" : "Based on the provided case study, here's an analysis of Elizabeth's situation and a suggested treatment plan:\n\nAnalysis:\n\n1. Diagnosis: Elizabeth has been diagnosed with 22q11.2 deletion syndrome, which explains her velopharyngeal disorder, severe articulation disorder, and severe hypernasality.\n\n2. Strengths: Elizabeth has normal hearing, middle ear function, and overall development (except for communication skills). This provides a good foundation for intervention.\n\n3. Challenges: The severe articulation disorder and hypernasality significantly impact her speech intelligibility. The velopharyngeal disorder affects her ability to produce normal speech sounds.\n\n4. Considerations: The 22q11.2 deletion syndrome may have additional implications for Elizabeth's long-term health and development, requiring ongoing monitoring and potential interventions in other areas.\n\nSuggested Treatment Plan:\n\n1. Speech-Language Therapy:\n   - Intensive articulation therapy focusing on improving speech sound production\n   - Exercises to strengthen velopharyngeal closure\n   - Resonance therapy to address hypernasality\n   - Language stimulation activities to support overall communication development\n\n2. Surgical Intervention:\n   - Consult with the surgeon about the potential benefits of pharyngeal flap surgery to improve velopharyngeal closure and reduce hypernasality\n\n3. Ongoing Medical Management:\n   - Regular check-ups with the geneticist to monitor for other potential issues related to 22q11.2 deletion syndrome\n   - Continued audiological monitoring to ensure hearing remains normal\n   - ENT follow-ups to manage any recurrent ear infections\n\n4. Psychological Support:\n   - Regular assessments to monitor cognitive and social-emotional development\n   - Potential play therapy or social skills training if needed\n\n5. Family Support and Education:\n   - Provide education to parents about 22q11.2 deletion syndrome and its implications\n   - Offer resources and support groups for families dealing with this genetic condition\n\n6. Educational Planning:\n   - Collaborate with Elizabeth's preschool or future school to ensure appropriate accommodations and support for her communication needs\n\n7. Interprofessional Coordination:\n   - Regular team meetings to discuss Elizabeth's progress and adjust the treatment plan as needed\n   - Ensure clear communication channels between all professionals involved in her care\n\n8. Long-term Monitoring:\n   - Schedule regular follow-up appointments with the cleft lip and palate clinic team to assess progress and make necessary adjustments to the treatment plan\n\nThis comprehensive treatment plan addresses Elizabeth's immediate speech and language needs while also considering the broader implications of her genetic condition. The interprofessional approach ensures that all aspects of her care are coordinated and tailored to her specific needs."
```

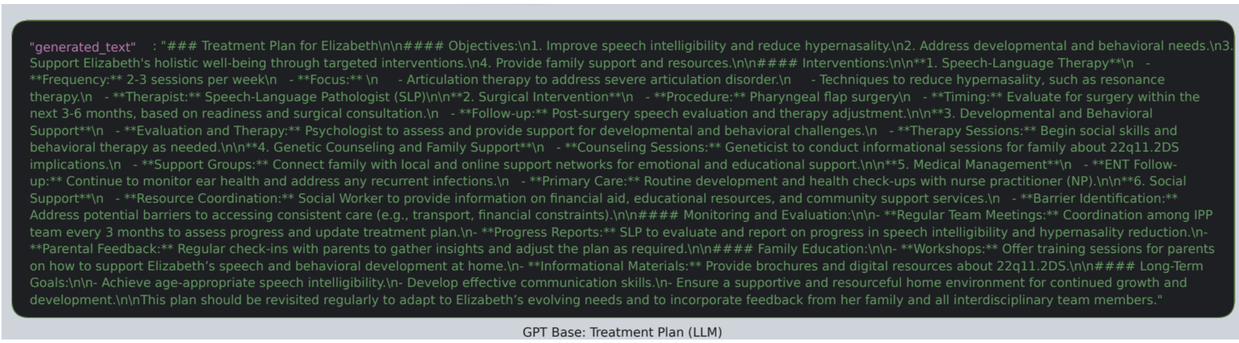

Claude Base: Treatment Plan (LLM)

Figure 20 Claude 3.5


```
"generated_text" : "Based on the provided case study, here is a comprehensive treatment plan with specific recommendations from each team member's perspective:\n\nAudiologist: Regular hearing screenings every 6 months to monitor for potential hearing loss associated with 22q11.2 deletion syndrome. Provide education to parents on signs of hearing changes and the importance of consistent follow-ups.\n\nEar, nose, & throat specialist (ENT): Schedule follow-up appointments every 6 months to monitor middle ear status and airway function. Recommend prompt treatment of any ear infections to prevent potential complications. Collaborate with the plastic surgeon on pre-operative assessment for pharyngeal flap surgery.\n\nGeneticist: Provide genetic counseling to Elizabeth's family regarding 22q11.2 deletion syndrome, including inheritance patterns and potential health implications. Recommend screening for other associated conditions such as cardiac anomalies, immune deficiencies, and endocrine disorders. Schedule annual follow-ups to monitor for the development of syndrome-related issues.\n\nNurse practitioner (NP): Coordinate care among team members and act as a liaison between the family and medical team. Monitor Elizabeth's growth and development, with particular attention to areas commonly affected by 22q11.2 deletion syndrome. Provide education to the family on managing Elizabeth's condition and recognizing potential complications.\n\nPlastic surgeon: Plan and perform pharyngeal flap surgery to address Elizabeth's velopharyngeal insufficiency and reduce hypernasality. Conduct pre-operative assessment and post-operative follow-ups to ensure optimal outcomes. Collaborate closely with the SLP for pre- and post-surgical speech therapy planning.\n\nPsychologist: Conduct regular developmental and behavioral assessments to monitor for learning difficulties or behavioral issues associated with 22q11.2 deletion syndrome. Provide support and guidance to Elizabeth and her family in coping with the diagnosis and potential challenges. Recommend early intervention services if needed.\n\nSLP (team facilitator): Intensify speech therapy sessions to address severe articulation disorder and hypernasality. Develop a comprehensive treatment plan focusing on improving velopharyngeal function and articulation skills. Collaborate with the plastic surgeon on pre- and post-operative speech therapy strategies. Conduct regular assessments to track progress and adjust therapy goals as needed.\n\nSocial worker: Assess family dynamics and provide support in coping with Elizabeth's diagnosis. Identify potential barriers to care and assist in finding solutions. Connect the family with community resources, support groups, and educational materials related to 22q11.2 deletion syndrome. Provide information on financial assistance programs if needed for ongoing medical care and therapies."
```


Claude prompt-tuned: Treatment Plan (LLM)

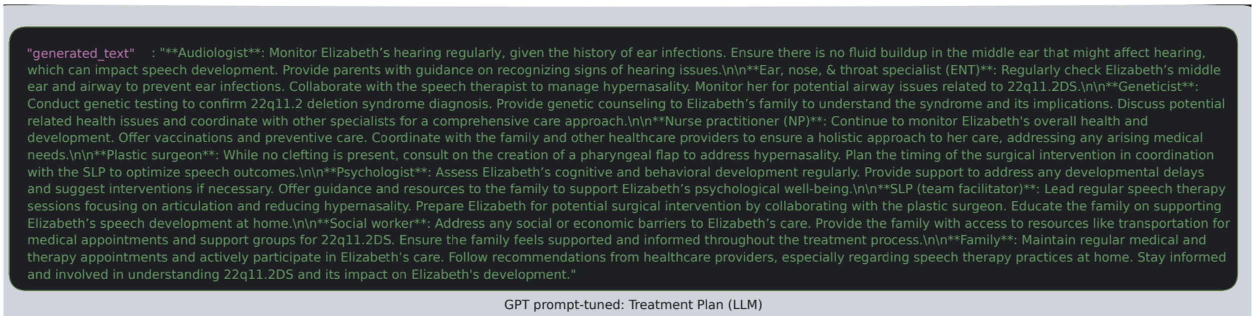
Figure 21 Claude 3.5



"generated_text" : "### Treatment Plan for Elizabeth\n\n#### Objectives:\n1. Improve speech intelligibility and reduce hypernasality.\n2. Address developmental and behavioral needs.\n3. Support Elizabeth's holistic well-being through targeted interventions.\n4. Provide family support and resources.\n\n#### Interventions:\n\n**1. Speech-Language Therapy**\n - **Frequency:** 2-3 sessions per week\n - **Focus:** \n - Articulation therapy to address severe articulation disorder.\n - Techniques to reduce hypernasality, such as resonance therapy.\n - **Therapist:** Speech-Language Pathologist (SLP)\n\n**2. Surgical Intervention**\n - **Procedure:** Pharyngeal flap surgery\n - **Timing:** Evaluate for surgery within the next 3-6 months, based on readiness and surgical consultation.\n - **Follow-up:** Post-surgery speech evaluation and therapy adjustment.\n\n**3. Developmental and Behavioral Support**\n - **Evaluation and Therapy:** Psychologist to assess and provide support for developmental and behavioral challenges.\n - **Therapy Sessions:** Begin social skills and behavioral therapy as needed.\n\n**4. Genetic Counseling and Family Support**\n - **Counseling Sessions:** Geneticist to conduct informational sessions for family about 22q11.2DS implications.\n - **Support Groups:** Connect family with local and online support networks for emotional and educational support.\n\n**5. Medical Management**\n - **ENT Follow-up:** Continue to monitor ear health and address any recurrent infections.\n - **Primary Care:** Routine development and health check-ups with nurse practitioner (NP).\n\n**6. Social Support**\n - **Resource Coordination:** Social Worker to provide information on financial aid, educational resources, and community support services.\n - **Barrier Identification:** Address potential barriers to accessing consistent care (e.g., transport, financial constraints).\n\n#### Monitoring and Evaluation:\n\n- **Regular Team Meetings:** Coordination among IPP team every 3 months to assess progress and update treatment plan.\n- **Progress Reports:** SLP to evaluate and report on progress in speech intelligibility and hypernasality reduction.\n- **Parental Feedback:** Regular check-ins with parents to gather insights and adjust the plan as required.\n\n#### Family Education:\n\n- **Workshops:** Offer training sessions for parents on how to support Elizabeth’s speech and behavioral development at home.\n- **Informational Materials:** Provide brochures and digital resources about 22q11.2DS.\n\n#### Long-Term Goals:\n\n- Achieve age-appropriate speech intelligibility.\n- Develop effective communication skills.\n- Ensure a supportive and resourceful home environment for continued growth and development.\n\nThis plan should be revisited regularly to adapt to Elizabeth’s evolving needs and to incorporate feedback from her family and all interdisciplinary team members."

GPT Base: Treatment Plan (LLM)

Figure 22 GPT-4o



"generated_text" : "**Audiologist**: Monitor Elizabeth’s hearing regularly, given the history of ear infections. Ensure there is no fluid buildup in the middle ear that might affect hearing, which can impact speech development. Provide parents with guidance on recognizing signs of hearing issues.\n\n**Ear, nose, & throat specialist (ENT)**: Regularly check Elizabeth’s middle ear and airway to prevent ear infections. Collaborate with the speech therapist to manage hypernasality. Monitor her for potential airway issues related to 22q11.2DS.\n\n**Geneticist**: Conduct genetic testing to confirm 22q11.2 deletion syndrome diagnosis. Provide genetic counseling to Elizabeth’s family to understand the syndrome and its implications. Discuss potential related health issues and coordinate with other specialists for a comprehensive care approach.\n\n**Nurse practitioner (NP)**: Continue to monitor Elizabeth's overall health and development. Offer vaccinations and preventive care. Coordinate with the family and other healthcare providers to ensure a holistic approach to her care, addressing any arising medical needs.\n\n**Plastic surgeon**: While no clefting is present, consult on the creation of a pharyngeal flap to address hypernasality. Plan the timing of the surgical intervention in coordination with the SLP to optimize speech outcomes.\n\n**Psychologist**: Assess Elizabeth’s cognitive and behavioral development regularly. Provide support to address any developmental delays and suggest interventions if necessary. Offer guidance and resources to the family to support Elizabeth’s psychological well-being.\n\n**SLP (team facilitator)**: Lead regular speech therapy sessions focusing on articulation and reducing hypernasality. Prepare Elizabeth for potential surgical intervention by collaborating with the plastic surgeon. Educate the family on supporting Elizabeth’s speech development at home.\n\n**Social worker**: Address any social or economic barriers to Elizabeth’s care. Provide the family with access to resources like transportation for medical appointments and support groups for 22q11.2DS. Ensure the family feels supported and informed throughout the treatment process.\n\n**Family**: Maintain regular medical and therapy appointments and actively participate in Elizabeth's care. Follow recommendations from healthcare providers, especially regarding speech therapy practices at home. Stay informed and involved in understanding 22q11.2DS and its impact on Elizabeth's development."

GPT prompt-tuned: Treatment Plan (LLM)

Figure 23 GPT-4o

Across all three models—LLaMA 3.2, Claude 3.5, and GPT-4o—we observed a consistent progression in treatment plan quality as we moved from the base prompt to prompt-tuned and MediSyn configurations except Llama whose prompt-tuned model gave the best results. Base model outputs, while often medically relevant, lacked consistent role attribution and structural clarity. Prompt-tuned outputs showed substantial improvement in aligning content with team member responsibilities, reinforcing interprofessional dynamics. The MediSyn outputs further refined this by synthesizing role-specific insights into highly organized and readable plans, frequently resembling real-world clinical documentation formats.

Among the models, Claude 3.5 consistently demonstrated strong formatting and fluency, especially in the prompt-tuned and MediSyn configurations. LLaMA 3.2 offered the richest medical detail, though it was sometimes lverbose or redundant. GPT-4o, while more concise, showed strong consistency in the MediSyn framework. These findings validate our system’s architectural progression and suggest that both prompt engineering and multi-agent collaboration significantly enhance treatment plan generation when from manually prepared input.

When compared to the actual treatment plan, we find that the prompt-tuned and MediSyn outputs from all models were able to preserve the essential medical context and interprofessional intent, while offering additional elaborations or reformatting for improved clarity. In some cases—particularly with Claude and LLaMA—models introduced medically plausible but inferred recommendations not present in the original, suggesting a tendency toward constructive hallucination. Overall, the MediSyn outputs most closely mirror the collaborative spirit and structural fidelity of the source plan.

4.3.2 LLM prepared data

We now extend our treatment plan comparison to LLM-prepared inputs, replicating the same evaluation pipeline as with manually curated data. Each model was provided with its own auto-extracted JSON (Section 3.2.2), allowing us to assess full end-toend performance. Our goal here is to understand how the quality of upstream data extraction impacts the performance of downstream generation. We again compare the base, prompt-tuned, and MediSyn configurations for each model while focusing on structure, accuracy, interprofessional clarity, and consistency with the case context.

The LLaMA MediSyn output demonstrates clear role segmentation and cohesive structure, integrating clinical and behavioral recommendations. Although some sections are verbose, the plan retains a professional tone and consistent formatting despite upstream inconsistencies, showing MediSyn's resilience to noisy input. Compared to its manually-fed counterpart, this output performs slightly better in maintaining formatting and flow.

Claude's MediSyn plan remains highly granular and well-organized, with clear role attributions and actionable strategies across domains such as care coordination and education. Compared to its manual-data version, Claude's synthesis agent exhibits slightly tighter integration, likely reflecting its strength in smoothing redundancies across overlapping agent outputs.

The GPT MediSyn output again follows a team-specific structure, but with even greater interprofessional depth. It introduces less common roles such as Care Coordinator and Educational Specialist, signaling the system's ability to reason beyond the immediate JSON input and simulate a full collaborative environment. Recommendations show better integration across roles. For example, the SLP section mentions home-based practice and resonance correction, while the ENT and Audiologist jointly focus on managing ear infections and tympanometry. The Social Worker role is also well-developed, covering transportation and financial barriers, an important IPP element often omitted in standard outputs.

Across all three generation strategies we observed distinct behavioral patterns among GPT-4o, Claude 3.5, and LLaMA 3.2 when operating on LLM-prepared data. GPT-4o consistently produced outputs that were concise, coherent, and wellstructured, particularly benefiting from prompt tuning. Its MediSyn outputs demonstrated strong role segmentation, although occasionally limited by brevity in certain roles. Claude 3.5 tended to generate highly verbose and well-articulated outputs, often including empathetic tones and explanatory elaborations. Its prompt-tuned and MediSyn plans were rich in profession-specific language, though at times overly detailed or slightly redundant. Nonetheless, Claude excelled in simulating realistic interprofessional discourse. LLaMA 3.2 showed more variability. Its base model outputs were often simplistic or overly broad, reflecting a lower capacity for nuanced interprofessional synthesis. However, prompt-tuning and MediSyn, markedly improved the result, producing detailed and diverse role-specific insights.

Overall, prompt tuning enhanced structure and specificity for all models, while MediSyn added a layer of agent-level diversity that closely mirrors the collaborative nature of IPE. While model strengths varied, the comparative analysis affirms the value of integrating both structured input and role-driven generation strategies to simulate authentic team-based medical planning.

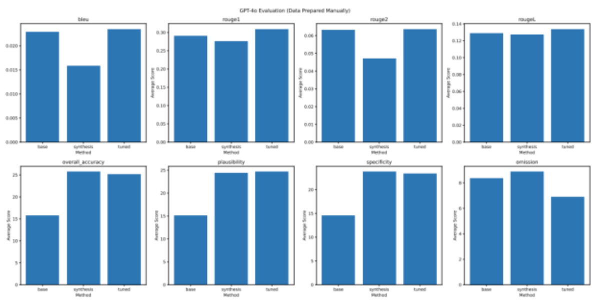
How does the quality of treatment plan generation differ when using manually prepared data versus LLM-prepared data?

While all models demonstrated the ability to generate reasonable treatment plans in both conditions, manually curated data consistently resulted in cleaner formatting, more precise role fidelity, and lower hallucination rates. By contrast, LLM-prepared inputs introduced slight drift in structure and completeness, likely due to minor extraction errors. Nevertheless, prompt tuning and MediSyn architectures proved robust, especially for Claude and GPT-4o, maintaining high-quality team-based plans even with imperfect input. Overall, while manually prepared data offers a slight edge in quality, the LLM-extracted data pipelines remain strong and practical for end-to-end simulation.

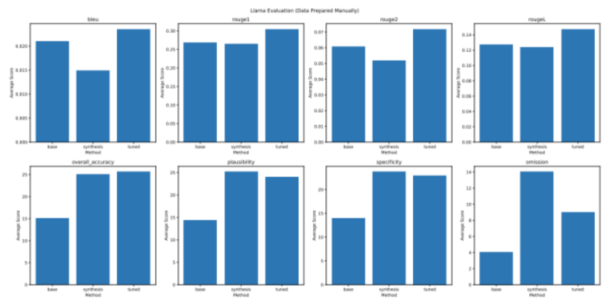
4.4 LLM vs. Prompt–Tuned LLM vs. MediSyn

We now compare three generation paradigms—Base Agent (single LLM), Prompt–Tuned, and the Synthesis-Agent variant of MediSyn—under two input conditions: (i) JSONs obtained by manual extraction, and (ii) JSONs obtained fully automatically by each LLM. Evaluation combines surface metrics (BLEU, ROUGE-1/-2/-L) with an LLM-as-a-Judge rubric (accuracy, plausibility, specificity, omission).

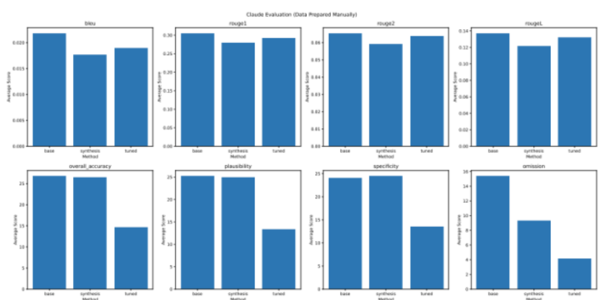
4.4.1 Manually prepared data



(a) GPT-4o



(b) LLaMA 3.2



(c) Claude 3.5

Figure 24: Metric-wise comparison across methods with manually curated data.

Across manually curated inputs, Claude 3.5 shows the strongest performance under the MediSyn configuration on human-aligned metrics such as accuracy, specificity, and plausibility. Although its base model achieves higher BLEU and ROUGE scores, these surface metrics do not reflect the interprofessional structure and role segmentation emphasized by MediSyn. For GPT-4o, prompt tuning leads across most metrics—particularly accuracy and plausibility—indicating the model's sensitivity to direct instruction. MediSyn remains close in performance but does not significantly surpass tuning. In the case of LLaMA 3.2, prompt-tuned output achieves the highest accuracy and plausibility, while MediSyn improves specificity at the cost of slightly higher omission and reduced BLEU/ROUGE alignment. Overall, the synthesis agent boosts interprofessional fidelity but introduces moderate tradeoffs depending on model characteristics.

4.4.2 LLM-prepared data

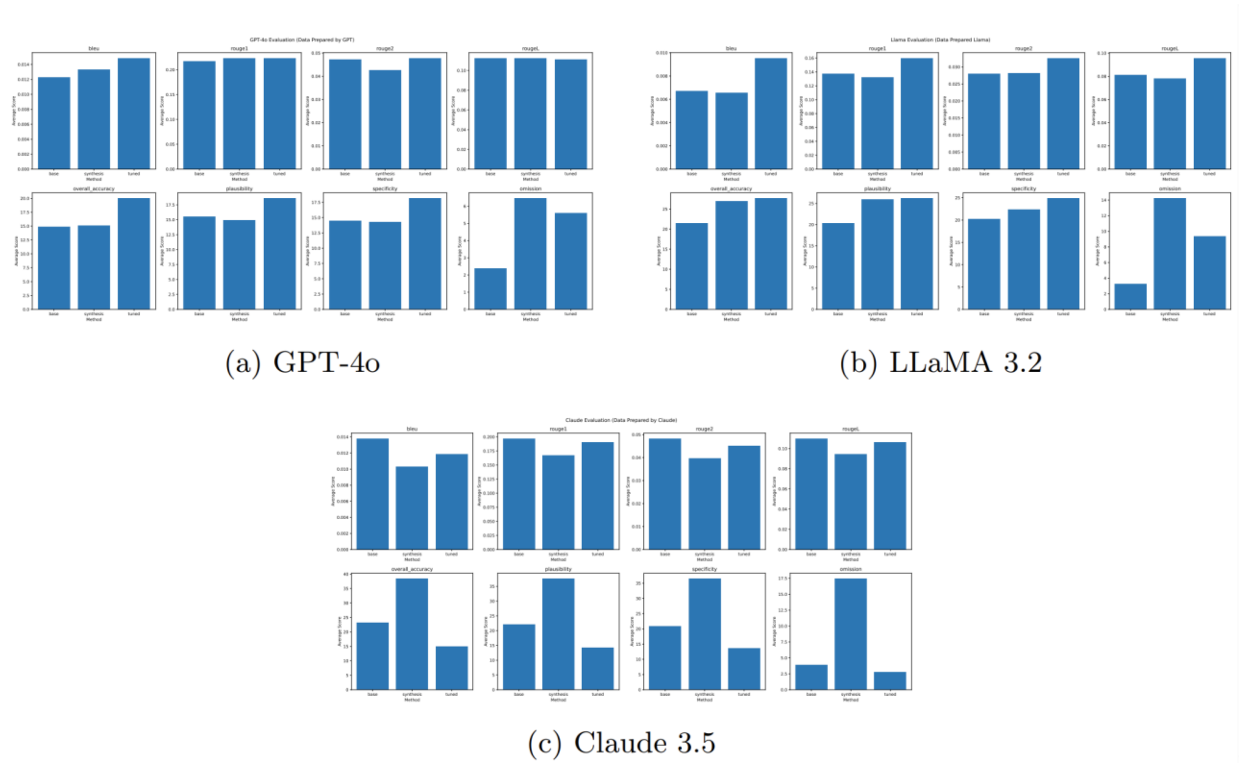
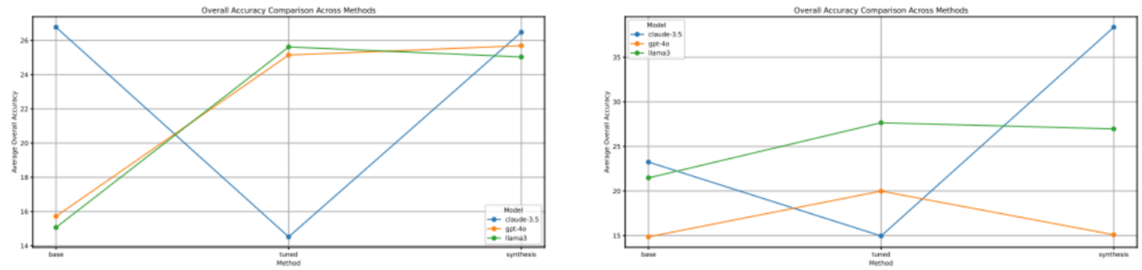


Figure 25:Metric-wise comparison across methods with LLM-prepared data.

LLM-extracted inputs introduce moderate noise, yet overall method rankings remain consistent across models. For Claude 3.5, the MediSyn variant yields the highest accuracy and specificity among the three configurations, demonstrating resilience to input imperfections and strong interprofessional structure. However, surface metrics (BLEU, ROUGE) remain highest in the base version, highlighting a tradeoff between lexical overlap and clinical structure. GPT-4o performs best under prompt tuning, showing peak values across accuracy, plausibility, and specificity, with relatively low omission. MediSyn remains competitive but does not surpass the tuned variant. For LLaMA 3.2, prompt tuning again leads in accuracy and plausibility, while MediSyn improves role segmentation and specificity, albeit with slightly elevated omission. These patterns affirm that prompt engineering remains effective under noisy conditions, while MediSyn enhances interprofessional fidelity and content depth across models.

These figures highlight that manual data consistently enhances absolute accuracy scores and reduces omissions across all models. Among the LLMs, Claude 3.5 benefits most from the MediSyn framework, especially under noisy input conditions—showing a pronounced increase in both accuracy and specificity. GPT-4o demonstrates its strongest performance with prompt-tuned inputs in both data regimes, while its MediSyn variant trails slightly, offering modest improvements in structural clarity but



(a) Manual JSON (b) LLM JSON

Figure 26:Average overall accuracy across models and methods.

lower overall accuracy. LLaMA 3.2 reaches peak accuracy with prompt tuning, though MediSyn improves specificity and maintains competitive results despite slightly elevated omission. Overall, MediSyn proves particularly effective for Claude and remains a resilient method under LLM-extracted data, reinforcing its utility in collaborative clinical reasoning contexts. For completeness, we include full metric tables below. These tables serve as a quantitative reference for the results summarized in the main text and figures.

	Base Agent			Prompt Tuned			Synthesis-Agent		
model	claude-3.5	gpt-4o	llama3	claude-3.5	gpt-4o	llama3	claude-3.5	gpt-4o	llama3
BLEU	0.021816491	0.022897534	0.020979131	0.018981152	0.023429586	0.02351592	0.017703129	0.015834182	0.014893625
ROUGE-1	0.305297053	0.290781431	0.268859856	0.291882742	0.30867233	0.304309476	0.279287886	0.27632991	0.265472946
ROUGE-2	0.065427345	0.063123451	0.060722117	0.064004364	0.063593493	0.071756048	0.059119787	0.047205562	0.05191511
ROUGE-L	0.13715102	0.12853165	0.126976533	0.132216424	0.133601063	0.147245837	0.121272152	0.127327397	0.123948985
Accuracy	26.771875	15.725	15.071875	14.521875	25.146875	25.61875	26.478125	25.690625	25.034375
Plausibility	25.225	15.05625	14.371875	13.38125	24.625	24.0125	24.940625	24.36875	25.140625
Specificity	24.034375	14.5125	13.96875	13.4125	23.35	22.921875	24.465625	23.73125	23.709375
Omission	15.378125	8.371875	4.078125	4.1375	6.871875	8.975	9.309375	8.859375	14.01875

Figure 27: Evaluation Table: Manually Prepared Data

	Base Agent			Prompt Tuned			Synthesis-Agent		
model	claude-3.5	gpt-4o	llama3	claude-3.5	gpt-4o	llama3	claude-3.5	gpt-4o	llama3
BLEU	0.013793611	0.012262974	0.006713744	0.011872966	0.014837258	0.009530648	0.010324213	0.013336691	0.006582385
ROUGE-1	0.197008489	0.217244534	0.137858245	0.191050421	0.223373087	0.159756379	0.167409889	0.223102102	0.132546991
ROUGE-2	0.048269877	0.047155786	0.028086315	0.04520276	0.047774683	0.032559656	0.039581989	0.042616133	0.028192104
ROUGE-L	0.109903807	0.112167282	0.081030912	0.106392056	0.111061832	0.095621644	0.094523526	0.112472864	0.078331159
Accuracy	23.24666667	14.85666667	21.47	14.95533333	19.98666667	27.65333333	38.37	15.09333333	26.96666667
Plausibility	22.10333333	15.48	20.26666667	14.24	18.56666667	26.18333333	37.55	14.94	25.96533333
Specificity	20.85666667	14.45333333	20.17666667	13.52	18.13533333	24.85333333	36.45333333	14.26	22.27333333
Omission	3.89	2.383333333	3.243333333	2.783333333	5.604666667	9.306666667	17.41666667	6.45	14.20666667

Figure 28: Evaluation Table: LLM Prepared Data

How does the quality of treatment plan generation differ when using manually prepared data versus LLM-prepared data?

Across all models and generation methods, manually curated input consistently led to higher accuracy, greater specificity, and fewer omissions. This is most pronounced for Claude 3.5, which saw substantial gains under the MediSyn framework when fed clean, manually structured JSON. LLaMA 3.2 also benefited from manual data, particularly in its prompt-tuned configuration, which achieved peak accuracy. GPT-4o showed smaller shifts across data regimes but remained most consistent with prompt tuning.

By contrast, LLM-prepared data introduced mild degradation in formatting content fidelity, likely due to field inconsistencies or paraphrasing during auto-extraction. However, MediSyn's synthesis step mitigated much of this noise for Claude and LLaMA, maintaining strong performance even with imperfect inputs. Prompt tuning helped GPT-4o maintain high plausibility and structure despite reduced granularity.

Overall, while manual input remains the gold standard for structured fidelity, the results confirm that LLM-generated input can produce high-quality treatment planning when paired with prompt tuning or multi-agent synthesis architectures like MediSyn.

5 Discussion

This work presents MediSyn, a modular multi-agent generation system designed to emulate real-world interprofessional collaboration in clinical decision-making. Evaluated across 15 diverse IPE case studies, our findings show that MediSyn consistently enhances the structural fidelity, role attribution, and contextual depth of treatment plans when compared to single-LLM and prompt-tuned approaches. The architecture's ability to simulate discipline-specific perspectives—while synthesizing them into 26 a unified treatment plan—proved especially effective for Claude 3.5 and LLaMA 3.2. Even GPT-4o, which tends to produce more compact outputs, demonstrated improved interprofessional coherence when integrated into the MediSyn framework.

A central theme in our analysis is the contrast between qualitative and quantitative evaluation outcomes. On the one hand, qualitative inspection—supported by examples in Section 4.3—clearly favored MediSyn for its role-sensitive structure, clarity, and alignment with IPE documentation practices. On the other hand, quantitative metrics such as BLEU and ROUGE did not always reflect these strengths, with prompt-tuned or even base configurations outperforming MediSyn on surface-level n-gram overlap. This divergence raises a critical question for the field: To what extent should we rely on traditional lexical metrics when evaluating clinical reasoning outputs, and when should human-aligned criteria—such as plausibility, specificity, and interprofessional clarity—take precedence? Addressing this requires deeper collaboration with domain experts to define evaluation benchmarks that align with both medical and educational standards.

Despite its promise, MediSyn has limitations. First, our use of GPT-4o as an LLMbased evaluator—while scalable and consistent—remains a proxy for expert human judgment. Although GPT-4o offers high fluency and general-purpose reasoning, its assessments may not fully capture clinical appropriateness or educational value. Future work should incorporate expert clinician evaluation for more robust validation. Second, while we employed 15 case studies covering a range of team compositions and conditions, this sample size remains modest for large-scale benchmarking.

Third, although MediSyn is designed to be model-agnostic, our analysis focused on three LLMs (GPT-4o, Claude 3.5, and LLaMA 3.2) without fine-tuning. Incorporating domain-adapted or instruction-tuned versions may reveal additional benefits or alter observed performance patterns.

Lastly, the modularity of MediSyn introduces its own challenges. Role prompts must be carefully crafted to ensure scope fidelity, and synthesis quality depends on the coherence of upstream agent outputs. Imperfections in auto-extracted input (e.g., ambiguous role mentions) can propagate into the final plan, especially under noisy LLM-prepared conditions.

Nonetheless, our results affirm that MediSyn provides a promising foundation for simulating collaborative clinical reasoning. Its ability to encode domain boundaries, promote inter-agent diversity, and synthesize profession-aware outputs makes it a valuable tool for both AI-driven medical education and the development of future interprofessional support systems.

6 Conclusion

We introduced MediSyn, a multi-agent generation framework that simulates interprofessional reasoning for medical treatment planning. By dynamically instantiating role-specific agents and synthesizing their outputs, MediSyn produces structured, collaborative plans aligned with IPE/IPP practices. Evaluated on 15 diverse case studies, MediSyn demonstrated strong performance in producing clinically coherent and role-attributed outputs, particularly when paired with models like Claude 3.5 27 and LLaMA 3.2. While prompt tuning improved alignment in many cases, MediSyn consistently enhanced structure and interprofessional clarity—especially under noisy, LLM-prepared inputs.

Our findings underscore the limitations of surface-level metrics and highlight the value of human-aligned evaluation criteria such as specificity and plausibility. As LLMs continue to evolve, frameworks like MediSyn offer a promising foundation for AIassisted education and team-based clinical reasoning, with opportunities for further refinement through expert evaluation and domain-specific tuning.

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